

GSFC NEON Series-1 CMO

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NS1-SMBA-SOW-0002, Revision B
Near Earth Orbit Network Series-1 Project, Code 473

**Near Earth Orbit Network
Project Series-1
Sounder for Microwave-Based Applications
Statement of Work**

**Near Earth Orbit Network Project Series-1
Sounder for Microwave-Based Applications
Statement of Work
Review/Signature/Approval Page**

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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

LEO/NEON Configuration Management Office
NASA/GSFC
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Change History Log

Revision	Effective Date	Description of Changes
Revision –	December 20, 2024	This version incorporates 470-CCR-24-0405 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0003 by the NS1 SMBA CCB on the effective date shown.
Revision A	April 17, 2025	This version incorporates NS1-CCR-0011 which was approved by the NS1 SMBA CCB on the effective date shown. Editorial change all instances of climate to weather.
Revision B	August 19, 2025	This version incorporates NS1-CCR-0013 which was approved by the NS1 SMBA CCB on the effective date shown.

Table of TBDs/TBRs/TBSs

Item No.	Location	Summary	Individual/ Organization	Due Date
1	IX.	Document numbers - The Contractor shall comply with all requirements in the PRD, NS1-SMBA-REQ-0001; CDRL, NS1-SMBA-CDRL-0001; ICD (TBD); and Mechanical Interface Control Document (MICD) (TBD).	Project	
2	1.1.2	Government Resident Office Support - The Contractor shall provide facilities to support nominally TBS Government on-site representatives at the instrument development/build site, including private securable office space, furniture, file/storage area, telephones, and printer or printer access, as well as access to a shared conference room from Contract award through satellite on-orbit acceptance (nominally Launch + 6 months).	Instrument Provider	
3	4.2.10.1	Test and Verification - The Contractor shall plan and implement the testing of instrument-spacecraft interfaces with the Spacecraft-Instrument Interface Simulator (SIIS) provided by the spacecraft contractor (TBR). Technical support will be provided by the spacecraft contractor at the instrument simulator provider facility. This will support early risk reduction of instrument-spacecraft interfaces.	Instrument Provider	
4	4.3.3	Ground Support Equipment - The Contractor shall provide two (TBR) additional sets of thermal control equipment for exclusive use at the spacecraft integration facility(ies).	Instrument Provider	
5	5.6.2	Instrument Calibration - The Contractor shall perform calibration at a minimum of three instrument temperatures representing the SMBA instrument low operational, mid operational, and high operational temperatures (TBS).	Instrument Provider	
6	5.6.2	Instrument Calibration - Intercalibration may be done at GSFC (TBR).	Instrument Provider	
7	6.5	Sensor Database Support - The Contractor shall supply XML (TBR) formatted	Instrument Provider	

		database inputs as required by the ground system.		
8	6.6.5.6 a)	Mission Rehearsals - Three (3) (TBR) instrument-focused rehearsals and dress rehearsals.	Instrument Provider	
9	6.6.5.6 b)	Three (3) (TBR) Support Operations Readiness Exercises as needed to prepare for and/or supplement Mission Rehearsals.	Instrument Provider	

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I. INTRODUCTION

The Sounder for Microwave-Based Applications (SMBA) is the backbone passive microwave instrument that produces global atmospheric temperature and moisture profiles for the National Oceanic and Atmospheric Administration (NOAA) and the science community. SMBA provides microwave data to support temperature and humidity sounding data in cloud covered and clear-sky conditions.

Risk Posture Statement

The Government intends to prioritize performance over process in accordance with the following risk posture:

*The SMBA instrument maintains **lower-risk tolerance** for delivering heritage-equivalent (including the Advanced Technology Microwave Sounder (ATMS), the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats, and Microwave Sounder (MWS)) performance and microwave sounding data continuity, for delivering hyperspectral, radio frequency interference (RFI) detection and intercalibration capabilities, for compliance with science performance requirements as specified in the Performance Requirements Document (PRD), and for adherence to safety standards. The SMBA instrument maintains **higher-risk tolerance** for adopting non-Government processes and standards that do not negatively impact areas of lower-risk tolerance to achieve cost-effective mission success. Where these areas conflict, achieving SMBA performance is most critical.*

This risk posture should be used for guidance in making risk-informed decisions.

This Statement of Work (SOW) covers the work required to develop, fabricate, integrate, test, analyze, verify, deliver, and provide post-launch support for the SMBA instruments required for the Near Earth Orbit Network (NEON) Program.

II. SCOPE

This SOW reflects the Contractor's scope within the period of performance as defined in the Contract. Throughout the remainder of the document, the instrument supplier will be referred to as the Contractor. The National Aeronautics and Space Administration (NASA), Goddard Space Flight Center (GSFC), and NOAA, including their support services Contractors, will be referred to as the Government. This SOW and the Contract Data Requirements List (CDRL) (NS1-SMBA-CDRL-0001) identify the deliverables associated with the SMBA instrument. The Contractor **shall** facilitate insight by the GSFC Program Office into the instrument design, development, integration and test progress and status.

This SOW defines the minimum effort required of the Contractor to design, develop, fabricate, integrate, test and deliver the SMBA instruments.

This SOW covers all SMBA instrument builds. Any requirement that is only applicable to the first flight unit or any other unit is specifically called out. Where no clarification regarding applicability is provided, the requirements are applicable to all flight units.

III. NEON PROGRAM AND NEON SERIES-1 PROJECT OVERVIEW

The NEON Program, a part of the National Environmental Satellite, Data, and Information Service Office of Low Earth Orbit (LEO) Observations, provides a framework to provide global measurements of Earth systems. These environmental measurements will be able to support a wide variety of atmospheric, terrestrial, marine, and polar observations. The data includes the numerical weather prediction models, fire and flood models, atmospheric chemistry observations, and multiple land imagery products that have been a crucial piece of the NOAA strategic goal to create a "Weather Ready Nation" and which provide essential information to NOAA's broader mission. Following the recommendations from the 2017 NOAA Satellite Observation System Architecture (NSOSA) study, the NEON Program intends to use a combination of small and medium satellite platforms in a disaggregated architecture. This approach will provide an efficient means of filling gaps quickly and taking advantage of emerging remote sensing technologies.

The NEON Series-1 Project is an initial operational project for the NEON Program featuring the development, integration, test, launch, and science operations of the SMBA instrument – a successor to the ATMS instrument, versions of which are on each of the Joint Polar Satellite System (JPSS) satellites. It will allow the NEON Program to provide continuity with the existing JPSS-era microwave sounder measurements as well as leverage newer technological advances to advance the collection of hyperspectral microwave sounding data that is crucial to numerical weather prediction models.

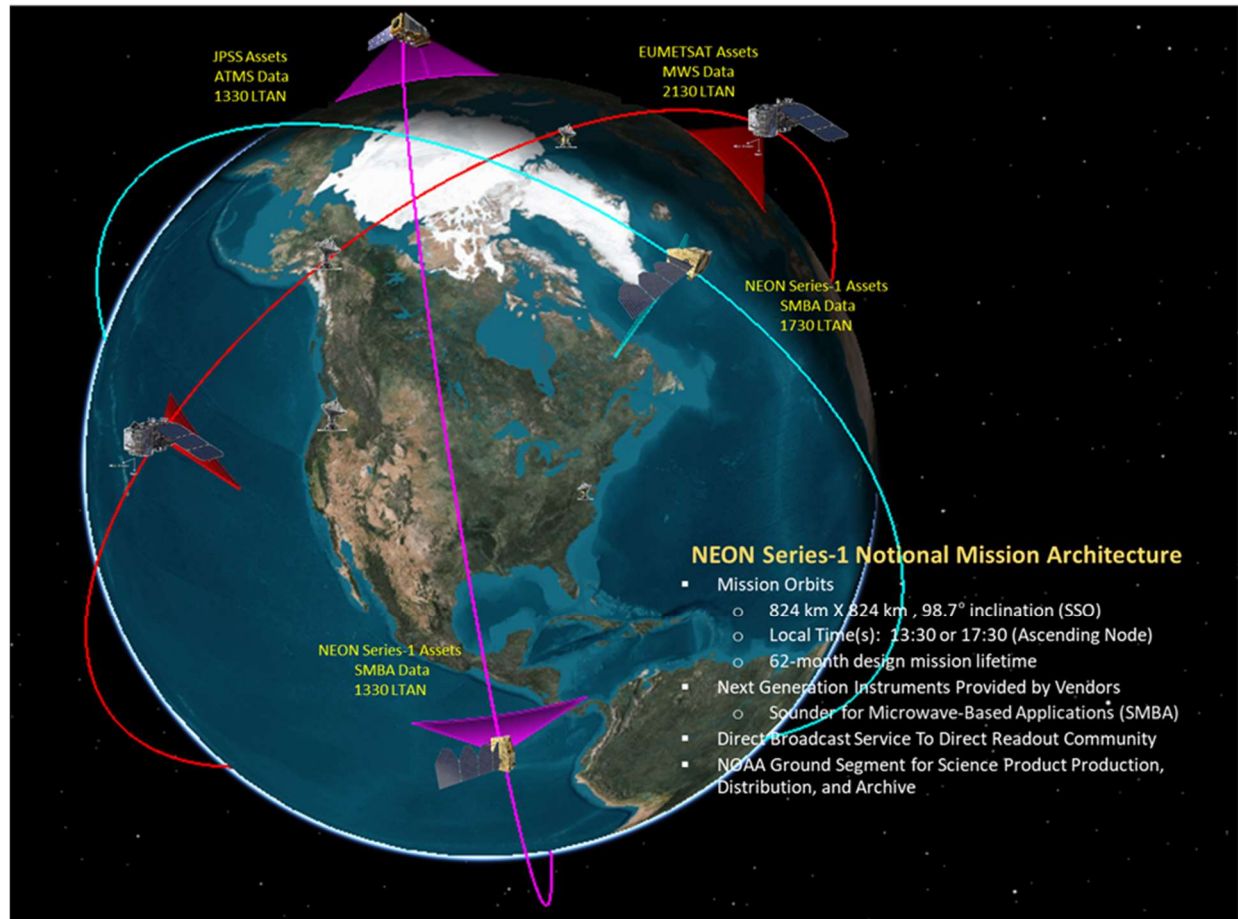


Figure III-1. Notional NEON Series-1 Project Architecture

Utilizing elements and best practices of the commercial space industry to meet Federal program level requirements, the NEON Series-1 Project will demonstrate that the NEON Program can provide operational hyperspectral microwave sounding observation capabilities with a small satellite that can be flexibly deployed to a variety of operational orbits. This first operational project, consisting of four individual flight missions to two distinct sun synchronous orbits (local time of ascending nodes (LTAN) of 1330 and 1730), will demonstrate the NEON disaggregated mission portfolio that provides the program with the flexibility of placing hyperspectral microwave sounders in either operational orbit (1330 or 1730 LTAN), at any desired launch cadence as the program needs to support hyperspectral microwave sounder needs.

The NEON Series-1 Project will also leverage mission assurance practices relative to established commercial missions and establish a referenceable knowledge base of rapid procurement practices that will benefit the future environmental observation missions planned in the high launch tempo NEON Program.

IV. THE SOUNDER FOR MICROWAVE-BASED APPLICATIONS

SMBA is the principal instrument aboard the NEON Series-1 satellite. The Instrument (hereafter referred to as the “SMBA instrument” or simply the “Instrument”), is a cross-track microwave sounder that collects hyperspectral microwave radiometric data that are used to calculate the vertical distribution of temperature (T), moisture (q), and pressure in the Earth’s atmosphere.

The instrument measures atmospheric radiances within the microwave range of approximately 20-300 GHz over a swath width of approximately 2,530 km from an altitude of 824 km.

Hyperspectral capability indicates fine spectral resolution. Fine spectral resolution is beneficial for both science and RFI detection capability. Hyperspectral microwave capability refers to microwave sounding systems with approximately 100 or more channels. The increased number of channels results in a finer spectral resolution than is achieved by traditional microwave sounding techniques over the same bandwidth.

Instrument data is acquired continuously, stored on-board the satellite, and subsequently downlinked to a ground network for capture, preprocessing, and routing to meteorological centers.

The instrument produces Raw Data Records (RDR) in an uncalibrated form. Engineering, instrument telemetry, science data, radiometric, and spectral calibration data is provided to enable the RDRs to be calibrated and geolocated. In addition, diagnostic RDR data may be acquired for testing purposes. The data are sent to the Ground System where they are used as input to science algorithms that generate Temperature Data Records, Sensor Data Records, and Environmental Data Records.

V. DEFINITIONS

Contractor – The developer of the item of reference. If the term Contractor is used in this document without specific reference to an item (e.g., SMBA, ground support equipment (GSE), etc.), then the term **shall** be interpreted to imply the SMBA Contractor. In all cases, the term Contractor also implies any and all associated suppliers and subcontractors. When used with a capital C, the term is interpreted to imply the SMBA Contractor.

Government – Federal Civil Servants and support Contractors (Contractors that represent NASA or are delegates for NASA).

Shall – Compliance by the Contractor is mandatory. Any deviations from these contractually imposed mandatory requirements require the approval of the Contracting Officer.

Should – Preferred good practices are identified by “should” statements. “Should” designates performance, which is requested, but not mandatory. Noncompliance of a “should” statement does not require Government approval but should be addressed at the earliest applicable project milestone.

May – At the discretion of the Contractor or Government.

Will – Designates the intent of the Government.

Applicable documents – The term “applicable documents” denotes Government-prepared and controlled documents and industry standards documents that levy additional requirements on the Contractor. The Contractor is required to comply with the documents, or the tailored version of the documents provided by the Government.

Reference documents – The term “reference documents” denotes documents that amplify or clarify the information presented in this document. The Contractor is not required to meet any requirements listed within these documents.

TBS – The term “to be supplied” (TBS) refers to a document or information that will be developed or provided.

TBR – The term “to be reviewed” (TBR) means that the stated requirements or values will be reviewed for appropriateness jointly by the Contractor and the Government and the information may be changed. The TBR requirement/value remains a requirement until formal process of configuration change review, approval, and Contract modification.

TBP – The term “to be proposed” (TBP) indicates required Contractor fill-ins as part of the solicitation process.

TBD – The term “to be determined” (TBD) identifies a missing requirement or value; indicates further research or analysis is needed to determine the information. All changes to resolve these items will be completed through a formal process of configuration change review, approval, and Contract modification.

Instrument – The SMBA flight hardware, support equipment (i.e., system test equipment (STE)/GSE), software, documentation, and/or data necessary to fulfill the requirements of the Contract and this SOW.

Satellite – The term “satellite” refers to the NEON Series-1 spacecraft bus integrated with the flight instrument(s). The term “satellite” is equivalent to the term “satellite” used in other NASA and NEON Program documents.

Virtual/Virtually – The terms “Virtual” or “Virtually” when applied to products being provided, indicates that the products will be provided electronically (e.g. via SharePoint, or Contractor portal).

Class I – A change that affects a configuration item's form, fit, or function. This could include changes to baselined documentation, technical requirements, Contract end items, interfaces, weight, power, data rate allocations, technical risks, or science performance. All proposed Class I changes require Configuration Control Board (CCB) approval.

Class II – A change that does not affect a configuration item's form, fit, or function. This could include changes to documentation only, minor hardware changes, or drawing changes that do not affect a baseline or interface. Class II changes do not require Project CCB review unless they will be written against Configuration Management (CM)-controlled documents.

VI. APPLICABLE DOCUMENTS

The Applicable Documents listed in this section apply directly to the performance of this SOW to the extent specified and invoked by this SOW. The Contractor **shall** comply with the documents to the extent specified herein. The Contract, Attachment N, provides the current issue revision and issue date as well as any additional Configuration Change Requests (CCR) for the applicable documents.

Document Number	Document Name
NS1-SMBA-CDRL-0001	SMBA Contract Data Requirements List (CDRL)
AS9100D	Quality Management Systems
NS1-SMBA-REQ-0001	SMBA Performance Requirements Document (PRD)
ISO9001:2015	Quality Management Systems
AS9003B	Inspection and Test Quality Systems Requirements for Aviation, Space, and Defense Organizations
NASA-STD-8739.4	Workmanship Standard for Crimping, Interconnecting Cables, Harnesses, and Wiring
IPC/WHMA-A-620x-S	Requirements and Acceptance for Cable and Wire Harness Assemblies
NASA-STD-8719.24 with Annex	NASA Payload Safety Requirements
GSFC-STD-1001	Criteria for Flight and Flight Support Systems Life Cycle Reviews
NASA-STD-8739.6	Implementation Requirements for NASA Workmanship Standards
IPC J-STD-001, class 3	Standard Soldering Requirements
GSFC-STD-6001	Area Array Package Assembly and Manufacturing Practices for Flight Hardware
IPC-2225	Sectional Design Standard for Organic Multichip Modules (MCM-L) and MCM-L Assemblies
ANSI/ESD S20.20	Protection of Electrical and Electronic Parts
IPC-6015	Qualification and Performance Specification for Organic Multichip Module (MCM-L) Mounting and Interconnecting Structures
IPC-6012	Qualification and performance of rigid printed boards
MIL-PRF-55110H	Performance Specification: Printed Wiring Board, Rigid General Specification
ECSS-Q-ST-70-60	Qualification and procurement of printed circuit boards
GEIA-STD-0005-1	Standard for Developing a Lead-Free Control Plan to Manage the Risks of Lead-Free Solders and Finishes in Aerospace, Defense, and High-Performance Soldered Electronic Products
GEIA-STD-0005-2	Standard for Mitigating the Effects of Tin Whiskers in Aerospace and High Performance Electronic Systems
NASA-STD-8739.12	Metrology & Calibration
GSFC EEE-INST-002	Instructions for Electrical, Electronic, and Electromechanical (EEE) Parts Selection, Screening, Qualification, and Derating
NASA-STD-6016	Standard Materials and Processes Requirements for Spacecraft

NASA-STD-6030	Additive Manufacturing Requirements for Crewed Spaceflight Systems Foundational Principles
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
NASA-STD 8719.14	Process for Limiting Orbital Debris

VII. REFERENCE DOCUMENTS

The documents listed in this section are those documents referenced in the SOW which contain information relating to the work required, but which do not constitute requirements.

Document Number	Document Name
AS9146	Foreign Object Damage (FOD) Prevention Program
NPR 8000.4	Risk Management Procedural Requirements
GSFC 500-PG-8700.2.8	FPGA Development Methodology
GSFC 500-PG-8700.2.7	Procedures and Guidelines for Design of Space Flight Field Programmable Gate Arrays
GSFC-STD-7000	General Environmental Verification Standard (GEVS)
NPR 7150.2	NASA Software Engineering Requirements
GPR 8700.6	Engineering Peer Reviews
GPR 8700.4	Integrated Independent Reviews
ANSI/EIA748	Industry Guide for Earned Value Management
IEST-STD-CC1246	Product Cleanliness Levels and Contamination Control Program
470-LEO-PROC-00010	Low Earth Orbit (LEO) Programs Division (LPD) Configuration Management Procedure
FAR Supplement Clause 1852.234-2	NASA Federal Acquisition Regulation (FAR) Supplement Clause 1852.234-2
NASA-STD-8739.8	Software Assurance and Software Safety Standard

VIII. PRECEDENCE

In the case of conflicting requirements, the order of precedence of documents not specifically called out in the Contract is: The Contract takes precedence over the SOW, followed by the CDRL, PRD, Interface Control Documents (ICD), and the remaining applicable documents. In the event of an inconsistency between any of the reference documents herein and the requirements of this SOW, the SOW **shall** take precedence. Any conflicts discovered by the parties between the requirements in this SOW and any referenced specification herein **shall** be brought to the attention of the other party for resolution.

IX. WORK TO BE PERFORMED

This section, along with the CDRL, GSFC NS1-SMBA-CDRL-0001, describes the specific work to be accomplished by the Contractor. In accordance with the requirements of this document, the Contract, all associated requirements documents, other attachments and applicable documents to this Contract, the Contractor **shall** provide the effort and resources that include, but are not limited to personnel, materials, equipment, and facilities necessary for the successful and on-time design, analysis, development, fabrication, assembly, integration, test, calibration, qualification, delivery, post-delivery support of the instrument.

The Contractor **shall** comply with all requirements in the PRD, NS1-SMBA-REQ-0001; CDRL, NS1-SMBA-CDRL-0001; ICD (**TBD**); and Mechanical Interface Control Document (MICD) (**TBD**). The Contractor is providing input to these documents in CDRL SE-8 and SE-9.

The Contractor **shall** employ contractor processes (Command Media) where practical.

The Contractor **shall** deliver to the Government instruments that are fully tested, qualified, and calibrated with verified compliance to the requirements of this Contract. The Contractor **shall** support the development of bilateral instrument documents, analyses, and models, which define the details of the instrument-to-spacecraft interface and instrument accommodation and operating information.

1 MANAGEMENT

1.1 Program Management

The Contractor **shall** appoint a dedicated Program Manager (PM) to direct and manage the SMBA instrument development who has responsibility for the overall technical performance, resource management, and schedule management and be responsible for effective and timely communication regarding program performance of the contractual effort and all subcontracts. The Contractor's designated PM **should** report to a level of company management appropriate to ensure prompt resolution of all problems. The Contracting Officer's Representative (COR) and Instrument Manager (IM) **will** maintain a close liaison with the Contractor PM.

The contractor **shall** use and maintain an Earned Value Management System (EVMS). The contractor's system **shall** use American National Standards Institute (ANSI)/Electronic Industries (EIA) 748, Industry Guide for Earned Value Management as a guideline. The EVMS **shall** be supported by the contractor's management processes and systems and in accordance with the Contractor's EVM processes.

1.1.1 Government Insight

To meet the mission and schedule requirements, the Government intends to establish a collaborative teaming environment with the Contractor to gain adequate insight of designs, Assembly, Integration, and Test activities, and operations. The Contractor's management, tracking, control, and reporting processes **shall** facilitate insight by the GSFC Program Office into the instrument progress and status. The Contractor **shall** provide the Government virtual access, on and off site, to the instrument, risk management, anomaly reporting, drawings, requirements verification, and planning and scheduling systems.

The Government **will** maintain a SharePoint, or similar, on-line collaborative space where the Contractor may share informal information with the Government.

The Contractor **shall** open Government attendance to Contractor and subcontractor/lower tier suppliers' internal data, reviews, audits, meetings, and other activities within the scope of the Contract where the Government has determined based on Contractor and/or subcontractor performance and/or risk to the program a need for such oversight. The Contractor **shall** notify the COR and the Government resident office representatives of meetings, reviews, or tests in sufficient time to permit meaningful Government participation.

The Contractor **shall** ensure that all software documentation, source code and software design models required for a NASA Software Independent Verification and Validation (IV&V) effort are made available to NASA IV&V personnel as required by the Government. The Contractor **shall** allow NASA IV&V review and participation before final product delivery to the Government.

1.1.2 Government Resident Office Support

To facilitate insight, the Government will establish a resident office that will include management, systems engineering, and quality assurance representation. The resident office representatives will attend the Contractor's daily meetings to stay current on activities and issues.

The Government will staff a contributive impactful resident office at the Contractor's facility to enable quick and efficient communication, problem resolution, and requirements clarification as necessary. The Government Resident Office will follow the Contractor's development and observe its processes including assembly, test, and inspections. The intent is to minimize overlaying Government processes and deliverables as much as possible with minimal disruptions to the development schedule. The Contractor **shall** provide facilities to support nominally **TBS** Government on-site representatives at the instrument development/build site, including private securable office space, furniture, file/storage area, telephones, and printer or printer access, as well as access to a shared conference room from Contract award through satellite on-orbit acceptance (nominally Launch + 6 months). The Contractor **shall** provide within these offices high-speed (broadband) internet access and periodic wireless access, as required, to an internet service provider within the Contractor's facility to allow for access to the GSFC network.

1.1.3 Reserved – Subcontract Management

The Contractor **shall** actively manage its Lower-Tier Suppliers (LTS) including robust mission and quality assurance and technical insight. The Contractor **shall** routinely verify LTS adherence to both Contractor-flowed down and approved subcontractor processes. The Government may require onsite personnel at LTS facilities in cases of frequent anomalies.

1.1.3.1 Incorporation of Lower-Tier Supplier Information

The Contractor **shall** provide to the Government the following LTS information at a minimum:

- a) A schedule of onsite visits to its key suppliers or those presenting risk to the program including planned and actual surveillance audits and be available for review at each Project Management Review (PMR).
- b) Earned Value Management System (EVMS) and financial data as required herein.
- c) Schedule and Schedule Reporting, typically via the Integrated Master Schedule (IMS).
- d) Project Status Reviews and/or Reports, typically via the PMR.
- e) Quality metrics as determined by the Prime or LTS typically summarized at the PMR as required.
- f) Design, Manufacturing, Integration and Test Review schedule, and contact information, minutes, or material discussed.
- g) LTS Risk items, risk mitigation tasks and status of risks **shall** be reported and included in each review if required and/or as requested by the Government.

1.1.3.2 Meetings with Lower-Tier Suppliers

The Contractor **shall** invite the Government to all formal/major reviews and acceptance activities at critical LTSs as negotiated between the Government and Contractor.

1.1.3.3 Flow Down of Requirements to Lower-Tier Suppliers

The Contractor **shall** flow all applicable requirements and controls stated or referenced herein to its LTSs. The LTSs **shall** supply a compliance matrix delineating their compliance with each of the requirements, provided to the Government upon request.

1.1.3.4 End Item Data Packages from Lower Tier Suppliers

The Contractor **shall** submit End Item Acceptance Data Packages (EIADP) from LTSs in accordance with CDRL MA-13.

1.2 Financial Management

The Contractor **shall** develop and provide Integrated Program Management Data Analysis Report (IPMDAR) CDRL PM-3. The Contractor **shall** use and maintain an EVMS in accordance with the EVMS requirements described in this Contract, under NASA Federal Acquisition Regulation Supplement Clause 1852.234-2. The EVMS **shall** be supported by the Contractor's management processes and systems and in accordance with the Contractor's EVM Plan. Reference the Contract Performance Report delivered as part of the Integrated Program Management Report (CDRL PM-3).

1.2.1 Performance Measurement Baseline

The Contractor **shall** define the Performance Measurement Baseline (PMB), completing a time-phased integrated plan to unify technical scope, schedule, and budget into an overall plan in accordance with the Contractor's EVM Implementation Plan. The Contractor **shall** define the work scope, establish a schedule for accomplishing the work, and estimate the resources required to accomplish the work. The Government will complete a PMB cost and schedule validation as required.

1.2.2 Integrated Master Schedule

The Contractor **shall** submit an IMS to be used to manage this instrument development effort. (Refer to CDRL PM-3.)

The Contractor **shall** submit and maintain a comprehensive listing of all external Giver/Receiver (G-R) items, also referred to as Receive-Deliver items to provide the project a description of each G-R indicated in the IMS. (Refer to CDRL PM-3.)

1.2.3 Change Control

In order to maintain baseline integrity and assure oversight of the baseline change control process, the Contractor **shall** use a formalized change control process for all Baseline Change Requests (BCR) via the Configuration Control Board process established by the Contractor. The Contractor **shall** provide notification, via e-mail submittal, of all BCRs over \$500,000 or BCRs that impact the critical path in the Integrated Master Schedule. Notification **shall** occur prior to implementation of the BCR into the baseline and annotate the type of change, Control Account Manager (CAM), Control Account impacted, identification of effects, the value of the change (if applicable), internal Contractor approval and the accounting month for which the BCR will be implemented.

1.2.4 Estimate at Completion

Once the PMB has been established and approved by the Government, a comprehensive Estimate at Completion **shall** be developed by the Contractor as part of the recurring detailed budget and planning process.

1.3 Risk Management

1.3.1 Reserved

The Contractor **shall** establish and maintain a comprehensive risk management program that meets the requirements of the Contractor's established Risk Management processes or NASA Procedural Requirement (NPR) 8000.4, Risk Management Procedural Requirements. All CCB authorized and funded risk burn-down (mitigation) activities **shall** be integrated into the IMS.

The Contractor **shall** invite the Government to attend Contractor Risk Management Board meetings. The Contractor **shall** maintain a list of risks from the risk board, present open risks at each PMR, and provide risk data (status, mitigations, descriptions, issues, and opportunities, etc.) to the Government upon request. SMA activities should be tightly linked with the project's Risk Management processes. For example, risks that evolve from reliability analyses that affect overall mission objectives should be managed in the project's risk database when not eliminated or mitigated to noncredible likelihood levels.

1.4 Configuration Management

The Contractor **shall** implement a configuration control process that enables the Contractor PM to assess and control configuration planning, identification, control, status accounting, and verification. The deployed CM processes should provide the means by which the Contractor is able to effectively assess and control all proposed changes to program or project documentation (contracts, statements of work (SOW), requirements, specifications, plans, procedures, drawings, hardware and software, etc.) that may affect form, fit, function, cost, or schedule.

The Contractor **shall** submit a waiver or deviation request for any non-compliance to any requirement. Any flight item that is found to be non-compliant may be re-worked or replaced with a compliant item.

The Contractor **shall** prepare and provide, as a minimum, the following configuration control documentation:

- a) CCB status reported at the Monthly Program Management Review in accordance with CDRL PM-1.
- b) Engineering Drawings in accordance with CDRL SE-3.
- c) The Configuration Item Identification List and the Computer Software Configuration Items (CSCI) in accordance with CDRL SW-1, SW-2, and RE-10.

1.4.1 Configuration/Change Control Board

The Contractor **shall** be a member of the SMBA GSFC CCB and required to participate in the review of project-level documents which are applicable to the Instrument Contractor (i.e., plans, specs, support of ICD updates, satellite integration and test (I&T) documents, etc.). The Government **shall** be invited to all Contractor Instrument CCBs. Government approval is required prior to the implementation of any Class I change. For Class I and II definitions, see 470-LEO-PROC-00010, LEO Programs Division (LPD) Configuration Management Procedure. Class I changes are modifications affecting the following:

- a) Contractually specified form, fit, and function.
- b) Government Furnished Equipment.
- c) Cost to the Government.
- d) Contractual deliveries.
- e) Schedule Contract milestones.
- f) Interchangeability of any item/unit that has already been delivered (excludes launched hardware).

All other changes are considered Class II or lower and may be incorporated without Government authorization. The Contractor **shall** notify the Government of change classifications and provide

all reasonable evidence required in support of change classifications. The Government reserves the right to upgrade any change classification to Class I if the Government determines the change to be improperly classified according to the Class I/Class II definitions above.

1.4.1.1 Design Changes

The Contractor **shall** document and control engineering changes as either Class I or Class II. CCRs **shall** be forwarded to the Government in accordance with CDRL SE-1, Engineering Change Requests.

1.4.1.2 Drawings and Drawing List

The Contractor **shall** prepare, maintain, and make available to the Government an Indentured Drawing List, a complete set of flight equipment drawings, and drawings for GSE at the Contractor's facility for each hardware type. (CDRL SE-3)

1.5 Reviews and Meetings

The Contractor **shall** conduct, host, prepare, and present instrument reviews and provide review packages in accordance with the stated CDRL/Data Item Description (NS1-SMBA-CDRL-0001) for each sensor Instrument as shown in Table 1-1.

A top-level overview of the reviews to be conducted is provided in Table 1-1 below. Further details for each review can be found in the subsequent sections of this SOW and the data pack requirements are provided in the CDRL.

Reviews **shall** be considered complete and approved by the Government when all outstanding action items are closed or a detailed plan for closure is submitted by the Contractor and approved by the Government.

Table 1-1. Top-Level Instrument Reviews

Review	Applicability	CDRL Reference	Chair
Integrated Baseline Review (IBR)	FM1	RE-8	Government
System Requirements Review (SRR)	FM1*	RE-3	Government
System Definition Review (SDR)	FM1*	RE-3	Government
Preliminary Design Review (PDR)	FM1*	RE-4	Government
ETU Test Summary Review	ETU	RE-14	Contractor
Critical Design Review (CDR)	FM1*	RE-5	Government

Review	Applicability	CDRL Reference	Chair
Manufacturing Readiness Review (MRR)	ETU FM1	RE-10	Contractor
Pre-Environmental Review (PER)	All Flight Instruments	RE-6	Contractor
Pre-Ship Review (PSR)	All Flight Instruments	RE-7	Government

***If any changes are made to the instrument (design, processes, procedures, etc.), this review may be required for any/all flight model(s) at the Government's discretion, depending on the magnitude of the changes.**

Unless stipulated otherwise or agreed to by the Government, the Contractor **shall** hold all Instrument Project Reviews and meetings at the Contractor's facility.

All reviews and meetings **will** provide a virtual attendance option.

1.5.1 Contractor-Led Meetings and Reviews

The Contractor **shall** implement a thorough technical peer review process of component and subsystem designs, conduct independent technical assessments of instrument activities, and hold formal design reviews. The reviews listed in this section should not be considered a complete set of reviews for the Contractor's program. Additional reviews that the Contractor deems necessary to successfully execute the program will be conducted at the Contractor's discretion.

The Contractor **shall** record and deliver meeting minutes for the meetings and reviews called out in this section.

1.5.1.1 Kickoff Meeting

The Contractor **shall** hold a Phase I Kickoff Meeting (RE-1) approximately two weeks after the Contract award.

The Contractor **shall** hold a Phase II Kickoff Meeting (RE-2) approximately two weeks after the Contract award.

The kickoff meeting agenda **shall** be developed jointly with the Government in accordance with CDRL RE-1.

The kickoff meeting will nominally be held at the vendor facility. Assumed duration is one day.

1.5.1.2 Program Management Review

The Contractor **shall** hold monthly PMRs in accordance with CDRL PM-1. These reviews will generally be held at the Contractor's facility, but may be held at the Government's facility, with mutual agreement between the Government and Contractor.

The PMR **shall** be designated to provide a management-level review of technical, schedule, and programmatic activities. A PMR will typically consist of a meeting to review formal status and splinter meetings to address special topics to be mutually determined prior to each review. The addition of special topics will be coordinated between the Government and Contractor Managers.

1.5.1.3 Integrated Baseline Review

The Contractor **shall** conduct an Integrated Baseline Review (IBR) to assess and validate the PMB in accordance with CDRL RE-8. This review will be scheduled as early as practicable and conducted in accordance with NASA requirements within 180 days of Contract award or formal Contract definitization. Additional baseline reviews may be required when substantial changes are made to the baseline (e.g., subsequent detailed planning, including routine rolling wave planning and program re-planning resulting from incorporation of significant Contract modifications). The IBR will be tailored to meet the needs of the NEON Program.

The Government will coordinate the timeframe and scope of the IBR with the Contractor. The IBR will ensure that the PMB captures the entire scope of work, is consistent with the Contract schedule requirements and has an adequate resource plan.

1.5.1.4 ETU Test Summary Review

The Contractor **shall** hold an ETU Test Summary Review to present the actual testing conducted compared to the plan, the results and any lessons of that testing, and any subsequent changes, either to the ETU and/or the flight design.

The ETU Test Summary Review may be combined with the Critical Design Review (CDR) if mutually agreed by the Contractor and Government.

The Contractor **shall** deliver the ETU Test Summary Review data package in accordance with CDRL RE-14.

1.5.1.5 Design Modification Reviews

Whenever there is a Contractor driven design change to the flight instrument that the Government feels cannot be adequately reviewed by other means, the Contractor **shall** hold a Design Modification Review (DMR) (CDRL RE-11) to review these changes at the Government's request. The Contractor **shall** include Government representatives on the review board.

1.5.1.6 Weekly Meetings

In order to track status, coordinate activities, and resolve issues, the Contractor **shall** support weekly meetings and telecons with the Government at mutually agreed to times. At a minimum, the weekly meetings should include: 1) a meeting to review status, work completed in the prior week, and a look ahead; 2) a technical issues meeting to cover technical issues as mutually agreed by the Contractor and Government; and 3) a brief weekly financial and business meeting to review staffing, spending, and funding, and special studies task status and hours.

1.5.1.7 Engineering Peer Reviews

The purpose of an Engineering Peer Reviews (EPR) is to reduce risk through expert knowledge infusion, confirmation of approach, and specific recommendations and document the peer

reviews in accordance with CDRL RE-9, EPR Data Packages. The Contractor **shall** chair and host EPRs at the Contractor's facilities with the Government invited to attend. The Contractor **shall** conduct multiple peer reviews, as appropriate and using GPR 8700.6 EPR as a guideline, over the life cycle of each subsystem and component with content consistent with the evolving design and development.

At a minimum, the Contractor **shall** complete a comprehensive series of subsystem or lower-level peer reviews prior to any instrument design reviews. Successful completion of these EPRs and resolution of associated technical issues and actions is an important aspect of entry criteria in the formal review process.

1.5.1.8 Manufacturing Readiness Review

The MRRs purpose is to verify the engineering baseline by a design, drawing, and engineering requirement review to assure that all prerequisites have been met, all necessary paperwork is correct and complete, and the final and correct material is released to manufacturing and that the manufacturing planning is complete and reflects the final configuration.

The ETU's MRR **shall** occur prior to assembly and CDR.

The Contractor **shall** prepare for and conduct an MRR CDRL RE-10 to verify the engineering baseline by a design, drawing, and engineering requirement review to assure that the release to manufacturing represents the final and correct flight equipment and that the manufacturing planning is complete and reflects the SMBA instrument configuration.

The approach to obviate GSE damage to flight hardware **shall** be presented prior to the start of testing and at subsequent review milestones.

Prior to performing work on flight hardware, the contractor **shall** identify a list of items, if any, that are sensitive to normal handling environments, such as presence of light, presence of metal objects (magnets), or presence of humidity outside of normal cleanroom and workmanship limits.

The MRR **shall** include manufactured Contractor and LTS components, sub-assemblies, and assemblies prior to the start of flight hardware manufacturing. These reviews will be conducted at EPRs. The ETU MRR is considered complete and approved by the Government when all outstanding action items are closed or a detailed plan for closure is submitted by the Contractor and approved by the Government.

1.5.1.9 Test Readiness Reviews

The Contractor **shall** conduct Test Readiness Reviews (TRR) in accordance with CDRL RE-13 prior to the start of any test of critical sub-assemblies, subsystems, assemblies, and the instrument.

The Contractor **shall** include Government personnel in TRRs and in the test procedure review process and provide a copy of the final procedure (CDRL IT-3).

The approach to obviate GSE damage to flight hardware **shall** be presented prior to the start of testing and at subsequent review milestones.

Prior to performing work on flight hardware, the contractor **shall** identify a list of items, if any, that are sensitive to normal handling environments, such as presence of light, presence of metal objects (magnets), or presence of humidity outside of normal cleanroom and workmanship limits.

The Contractor **shall** deliver TRR Data Packages in accordance with CDRL RE-13.

1.5.1.10 Test Data Reviews

The Contractor **shall** conduct Test Data Reviews (TDR) for data collected from instrument sub-assembly, subcontracted subassembly, and instrument, as well as prior to the shipment of deliverable GSE.

The Contractor **shall** include Government personnel in TDRs.

The Contractor **shall** conduct a TDR of the interim data during the qualification or acceptance test process of the instrument prior to the breakdown of any test setup (e.g., vibration testing, Electromagnetic Capability (EMC)/Electromagnetic Interference (EMI) testing, thermal vacuum (TVAC) testing, etc.) to ensure successful collection of required data, and prior to the Pre-Ship Review (PSR), to determine readiness.

1.5.1.11 Software Reviews

The Contractor **shall** plan and conduct software reviews in accordance with the Instrument Software Development Plan, CDRL SW-1.

Unless stipulated otherwise by the Government, the Contractor **shall** invite Government personnel to software reviews.

The Contractor **shall** conduct, at a minimum, the following software EPR:

- a) Software Requirements Specification Review/Walkthrough
- b) Software Preliminary Design Review
- c) Software Critical Design Review
- d) Software Test Readiness Review
- e) Software Qualification Review
- f) Software Test Readiness Review (SWTRR)

The Contractor **shall** document the scope and process for conducting software peer reviews in the Instrument Software Development Plan in accordance with CDRL SW-1.

1.5.1.12 Technical Interchange Meetings and Other Reviews and Meetings

A Technical Interchange Meeting (TIM) is an informal meeting between the Contractor and Government representatives to, for example, reach an understanding of an operation or analysis, presentation of test results, discuss planned interface changes, or plan for an upcoming test. TIMs typically are held at the Contractor's facility and typically run no more than two days.

Virtual attendance may be done if agreed upon by the Contractor and Government. Actions **shall** be tracked by the TIM organizer.

The Contractor **shall** conduct technical/management interchange meetings as needed to discuss and resolve any technical and/or managerial issues or concerns.

The Contractor **shall** invite the Government as a participating member in the Contractor's Anomaly Review Board (ARB), Failure Review Board (FRB), Material Review Board (MRB), Parts Control Board (PCB), and Materials and Processes Control Board.

In addition to the meetings and reviews described above, the Contractor **shall** support routine informal meetings and teleconferences with the Government.

1.5.1.13 LTS and Subsystem Reviews

The Contractor **shall** plan and conduct subcontract and subsystem reviews.

Unless stipulated otherwise by the Government, the Contractor **shall** invite Government personnel to subcontract and subsystem reviews.

The Contractor **shall** conduct Preliminary Design Reviews (PDR) and CDRs for all major subsystems whether the subsystems are provided by the Contractor or by a subcontractor.

Subcontract and subsystem reviews **shall** be conducted by the Contractor. The review board **shall** be chaired and staffed by Contractor-selected independent reviewers for the hardware under review.

The Contractor **shall** submit Subsystem/Subcontract Review Data Packages in accordance with CDRL PM-1.

1.5.2 Government-Chaired Instrument Reviews

The Government will chair the major milestone reviews in the instrument life cycle where information is formally presented to a panel of Government experts and external reviewers. These reviews will typically involve 10 to 50 Government representatives. The Contractor **shall** determine the optimum number of consecutive days to perform each of these reviews to meet the requirements in GSFC-STD-1001, but not to exceed four days. Formal action items are logged and tracked by the Government Review Team.

All major reviews will be convened, and review boards appointed and chaired by the Government. The Contractor **shall** demonstrate compliance with the review success criteria of GSFC-STD-1001, Criteria for Program Flight Critical Milestone Reviews, as applicable to spaceflight instruments. Non-compliance may result in a "delta" review at the discretion of the Government.

The Contractor **shall** develop and apply a process for capturing, responding, and closing assigned action items assigned by the Government; reviews are not complete until actions are complete or a detailed plan for closure is submitted by the Contractor and approved by the Government.

Where the Government determines an additional review is required, a “delta” review **shall** be conducted. This “delta” review **shall** meet the requirements of the appropriate system-level review (CDR, etc.) as defined herein and in the applicable CDRL. Once the Government has decided a “delta” review is necessary, the Government will formally inform the Contractor of the tailored scope of the delta review and the deadline to hold the review.

1.5.2.1 System Requirements/Definition Review

The Contractor **shall** conduct a System Requirements Review (SRR)/System Definition Review (SDR) to ensure proper and complete requirements definition and flowdown from the Level 3 requirements levied by the Government to lower levels.

The Contractor **shall** deliver an SRR/SDR Data Package in accordance with CDRL RE-3.

1.5.2.2 Preliminary Design Review

The Contractor **shall** conduct a PDR at the conclusion of the preliminary design. The Contractor may combine the PDR with the engineering development model Pre-Build Review pending concurrence from the Government IM and respective review chairpersons.

The Contractor **shall** deliver a PDR Data Package in accordance with CDRL RE-4.

1.5.2.3 Critical Design Review

The Contractor **shall** conduct a CDR prior to the start of procurement and/or manufacture of the FM-1 hardware unless the Contractor requests and the Government concurs with an earlier start.

The Contractor **shall** obtain Government concurrence to purchase any materials Flight Model (FM) materials prior to the CDR.

All technologies **shall** achieve a Technical Readiness Level (TRL) of TRL-6 by CDR.

The Contractor **shall** deliver a CDR Data Package in accordance with CDRL RE-5.

1.5.2.4 Pre-Environmental Review – Each Instrument

The Contractor **shall** conduct a Pre-Environmental Review (PER) prior to the start of environmental testing of the ETU and each FM instrument to establish the readiness of the system and to evaluate the environmental test plans and procedures.

The Contractor **shall** deliver a PER Data Package in accordance with CDRL RE-6.

1.5.2.5 Pre-Ship Review – Each Instrument

The Contractor **shall** conduct a PSR at the conclusion of the integration and test effort prior to shipment and/or storage of each SMBA FM. If the instrument is stored for a period greater than six months prior to shipment, the Contractor **shall** prepare an abbreviated (half-day) delta-PSR immediately prior to shipment to review logistical shipping details as well as any additional work or testing performed on the instrument since the PSR.

The Contractor **shall** deliver a PSR Data Package, in accordance with CDRL RE-7.

1.5.3 Higher-Level Reviews

The Contractor **shall** provide support to other reviews as required by the NEON Program. Once the instrument is delivered, the Contractor **shall** support all reviews necessary for the successful integration, test, launch, and operation of this instrument as defined in Section 6 of this SOW.

Note: These “other reviews” will be held at the spacecraft contractor, GSFC or launch facility.

1.5.3.1 Satellite/Mission-level Reviews

The Contractor **shall** provide technical support to all Satellite/Mission-level reviews, including Satellite integration and test reviews, conducted at the spacecraft contractor's facility.

The Contractor **shall** support the Satellite/Mission-level reviews by presentations and/or documentation; for example, by providing and/or presenting instrument overview or design summary, test, and data evaluation summaries, etc., in accordance with CDRL RE-15, Inputs to Mission-Level Reviews.

The Contractor **shall** support all Satellite-level Reviews including but not necessarily limited to:

- a) Integration Readiness Review (1 day for each FM)
- b) Satellite Pre-environmental Review (3 days for each FM)
- c) Satellite Pre-storage Review (3 days for each FM)
- d) Satellite Pre-ship Review (3 days for each FM)

1.5.3.2 Program-Level Reviews

The Contractor **shall** support the Program-level Reviews by presentations and/or documentation, for example, an instrument overview or design summary, test, and data evaluation summaries, etc.

The Contractor **shall** support the following Program-level Reviews:

- a) Mission SDR (3 days)
- b) Mission PDR (3 days)
- c) Mission CDR (3 days)
- d) System Integration Review (3 days for each FM)

Any inputs **shall** be provided in accordance with CDRL RE-15, Inputs to Higher Level Reviews.

1.5.3.3 Operations Reviews – Each Flight Model

The Contractor **shall** support the Operations Reviews for each FM by presentations and documentation; for example, by providing and/or presenting instrument overview or design summary, test, and data evaluation summaries, etc., in accordance with CDRL RE-15, Inputs to Higher Level Reviews.

The Contractor **shall** support the following Operations Reviews for each FM at the Satellite Operations Facility:

- a) Mission Operations Review (MOR) (3 days)
- b) Flight Operations Review (3 days)
- c) Operational Readiness Review (ORR) (3 days)
- d) Post-Launch Assessment Review (2 days)

1.6 Data Retention

The Contractor **shall** retain those data necessary to provide a technical and program database of significant decisions, plans, and the execution thereof as well as those data providing design, manufacturing, and test history of all deliverable items for the period of 10 years from the date of the Contract final payment/settlement. Data to be retained include the following:

- a) Part and material procurement: Test, screening, and inspection data, including nonconformance reports.
- b) Subassembly and higher assembly: Fabrication, assembly, test, and inspection data.
- c) Shop travelers (including travelers at lower tier suppliers).
- d) Kitting/traceability records.
- e) Executed test procedures.
- f) Listing of test equipment used.
- g) Test and calibration data including nonconformance reports.
- h) GSE certification, calibration of test equipment.
- i) As-built/as-designed verifications.
- j) Operating time logs.
- k) Total test time for each environment.
- l) End Item Data Packages.
- m) Design review data and packages.
- n) Drawings.
- o) Contractor Parts, Materials, and Processes Control Authority records of all performed testing:
 - Qualification and acceptance test data.
 - Destructive physical analysis reports with samples.
 - Radiation hardness assurance test data.
- p) Computer Aided Design data files that capture the mechanical design of the instrument and associated GSE.
- q) Photographic documentation.
- r) Models used to perform analyses for delivered CDRLs (e.g., thermal, optical, contamination, General Reflector Antenna Analysis Software Package).
- s) Environmental records.
- t) Retain numeric data in digital format.
- u) Retain or archive software lookup tables.

2 SYSTEMS ENGINEERING

The Contractor **shall** perform systems engineering to support all instrument-related activities during all stages of development. The systems engineering effort **shall** include, but is not limited to, analyses of technical requirements; functional and performance allocation of derived requirements; traceability, definition, and maintenance of all interfaces; instrument design and verification of all defined, allocated, and derived requirements; systems analyses and special studies as required; risk management support; and tradeoff analyses. This includes but is not limited to the following specific activities:

- a) Providing technical direction and oversight throughout all phases of the program.
- b) Leading, supporting and/or preparing documentation for all peer reviews as well as those reviews defined in Section 1.5.
- c) Performing coordination, studies and analyses required to interface the instrument to the spacecraft MICD, (ICD), including instrument specific support to ground system requirements development and operations.

2.1 Deviations and Waivers

Requests for deviations or waivers to the Government's requirements **shall** be submitted to the Government's Contracting Officer for Government approval and documented as Engineering Change Requests in accordance with CDRL SE-1.

2.2 Requirements Analyses, Derivations, and Allocations

The Contractor **shall** provide the definition, allocation, derivation, and traceability of system and subsystem requirements and the verification approach CDRL SE-4.

The Contractor **shall** conduct complete analyses and simulations in support of technical requirements compliance demonstrations to fully establish, define, maintain, and control allocations for all required performance and design parameters, including:

- a) Flow-down and traceability of Instrument requirements from SMBA requirements per CDRL SE-5 Instrument Performance Specification.
- b) Functional and performance allocations and derivations.
- c) Developing the Instrument Concept of Operations in accordance with CDRL SE-7.
- d) Preparing the Instrument Mass Properties Reports in accordance with CDRL SE-11.
- e) Developing the Beginning of Life (BOL) design performance requirements based on the PRD End of Life (EOL) requirements and documenting the supporting analyses in CDRL SE-5, Instrument Performance Specification.

If a Model-based systems engineering approach is used to support the requirements, design, analysis, verification, and validation associated with the development of the SMBA instrument models, any analysis and supporting documents (CDRL SE-6) **shall** be provided virtually.

2.2.1 Technical Performance Metrics

The Contractor **shall** implement and manage Technical Performance Metrics (TPM), comprised of key performance requirements for the end item product, including tracking and reporting TPM status and changes. The Contractor **shall** prepare and submit TPM status detailing current estimated value, requirement, margin, trend, and corrective action if required, in accordance with

CDRL PM-1. If there are no changes to report reference can be made to the last reported TPM baseline.

2.2.2 Instrument Interface

The Contractor **shall** support the development of bilateral instrument documents (Government to Contractor, Spacecraft to Instrument), as well as analyses and models which define the details of the instrument-to-spacecraft interface, instrument accommodation, and operating procedures and information for the instrument.

2.2.2.1 ICD & Mechanical ICD

The Instrument Contractor **shall** develop and deliver input to an Instrument ICD (SE-8) and MICD (SE-9) that includes the instrument's interface requirements that will be provided to the spacecraft contractor to form the basis of the instrument-to-spacecraft ICD/MICD. The Contractor **shall** collaboratively work with the Government and the spacecraft contractor to review and maintain accurate instrument-to-spacecraft ICD(s) and MICD(s) and recommend corresponding and/or similar changes to the PRD. The Contractor **shall** verify ICD requirements in accordance with CDRL SE-4.

The Instrument Contractor will work with future spacecraft contractor to implement an ethernet-type data interface between the instrument and spacecraft.

2.2.2.2 Giver/Receiver

The Contractor **shall** collaboratively support the development, maintenance, and review of the SMBA GR list of CDRLs and other items (such as, but not limited to, documents, hardware, and software) to be exchanged between the Contractor and other organizations. The Contractor **shall** provide the items documented in the GR list per the schedules documented and agreed to therein. The GR list **shall** document both items to be delivered to outside organizations (Givers) and items to be delivered from outside organizations (Receivers).

The GR List **shall** be delivered as incorporated into the IMS in accordance with CDRL PM-3, IPMDAR.

2.3 Contamination Control

The Contractor **shall** incorporate contamination control into the instrument design, processes, and planning that assures mission success per IEST-STD-CC1246, PRD Section 5.7.7, and in accordance with CDRL SE-8 and IT-4. The instrument **shall** be maintained at a cleanliness level as defined in SMBA Satellite Contamination/Cleanliness Control Plan. The Instrument **shall** be cleaned prior to its use, shipment, and integration onto the spacecraft. Vent locations and paths **shall** be defined in the MICD. The Contractor **shall** monitor the instrument for contamination effects and validate instrument contamination controls. The Contractor **shall** document contamination analyses as specified in the analysis reports in accordance with CDRL SE-6c.

Bake out **shall** be done prior to delivery, such that contamination requirements are met prior to delivery to the spacecraft facility to minimize contamination of other contamination sensitive components/subsystems on the space vehicle during storage and on orbit.

Multi-layer insulation blankets, harnesses and cables, and electronic boxes/modules to be included in TVAC testing **shall** be baked out separately from the instrument.

Purge port: If there is a conditional purge, it **shall** be delivered via a purge port rather than a feed through a bag.

For long term storage there **shall** be a purge port that remains capped for I&T. The purpose would be to provide flow through the instrument volume to prevent the build-up of corrosives due to offgassing.

If the Instrument does not have a purge port accessible after delivery and integration, the Contractor **shall** conduct a contamination study to determine the best value approach to protecting the instrument from contamination sources that exist through launch and deliver the data, results, and recommendations to the Government for disposition.

Powered operations while bagged: The Contractor **shall** verify whether the Instrument can operate while bagged and define how long it can operate while bagged in the case of thermal constraints.

The Contractor **shall** supply any instrument bags needed after delivery to the spacecraft.

Post-delivery support: The Contractor **shall** be responsible for instrument cleaning and sampling post-delivery through encapsulation.

The MGSE **shall** be delivered in a state that prevents particulate contamination.

The contractor **shall** implement a foreign object damage/debris (FOD) program to address the preservation of product with respect to FOD prevention using AS9146, FOD Prevention Program – Requirements for Aviation, Space, and Defense Organizations, as a guideline.

2.4 Design Engineering

The Contractor **shall** design and deliver an instrument that meets all requirements in the PRD, , CDRL, ICD and MICD. The Contractor **shall** develop and deliver the Instrument Performance Specification in accordance with CDRL SE-5. The Contractor **shall** make available upon request by the Government design specifications and drawings for all subcontracted and purchased items.

The Contractor **shall** conduct subsystem requirements and design EPRs in accordance with CDRL PM-2 and CDRL RE-9 for, at a minimum, the following subsystems prior to milestone reviews:

- a) Electrical Systems
- b) Antenna
- c) Scan Drive and electronics
- d) Receivers
- e) Hyperspectral Processing subsystem in accordance with SE-6e
- f) RFI Processing subsystem in accordance with SE-6f

- g) Thermal in accordance with SE-6b
- h) Structural in accordance with SE-6a
- i) Software in accordance SW-2
- j) Firmware in accordance SE-13

2.5 Design, Analysis, and Verification Documentation

The Contractor **shall** perform and document all analyses of the data and information from the design, qualification testing, acceptance testing, compatibility testing, and on-orbit testing of the Contractor's hardware and software which are required to ensure that the instrument will meet its specifications and objectives.

2.5.1 Design Analyses

The Contractor **shall** perform the necessary systems engineering analyses to support the satisfaction of the Contract requirements in accordance with CDRL SE-6, SE-5, and CV-4, including the following:

- a) Radiometric Math Model with uncertainties defined, explained, and quantified.
- b) Perform a flow-down of the performance requirements to meet its requirements with sufficient margin.
- c) Pointing knowledge and error budget of the entire instrument.
- d) Data system throughput, storage, and any playback analyses.
- e) Antenna cross-polarization characterization with respect to scan angle.
- f) Antenna alignment design, tolerances, and error budget.

All analyses performed by the Contractor under this Contract **shall** be made available to the Government upon request.

Design analyses **shall** be submitted in accordance with CDRL SE-6 in support of the design reviews. The Contractor may submit existing analyses for any unit type for which space qualification has been demonstrated to comparable requirements. Contractor format and analyses configuration is acceptable.

2.5.1.1 Structural/Dynamic Analyses and Verification Model and Plan

The Contractor **shall** perform comprehensive structural and mechanical analysis and deliver the results (CDRL SE-6a) documenting the analysis, and develop and deliver Structural and Dynamic Models and a Model Verification Plan in accordance with CDRL SE-6a.

2.5.1.2 Thermal Analyses and Modeling

The Contractor **shall** provide documented verification that the hardware meets all thermal design requirements for the program. The Contractor **shall** develop and deliver a comprehensive thermal analysis, a Thermal Analysis Report documenting the analysis, and Thermal Models in accordance with CDRL SE-6b. Thermal analyses **shall** be performed to demonstrate that test configurations and associated ground support equipment (e.g., cold plates, chillers, chambers, thermal targets) have sufficient control authority to TVAC cycle the flight hardware to protoflight levels and simulate mission environments during thermal balance tests. The detailed analysis **shall** demonstrate that all parts meet established thermal design criteria, based upon program de-rating or electrical performance requirements.

The instrument Contractor **shall** define temperature limits of components as well as test tolerances during PDR, in coordination with the Government responsible personnel.

Thermal verification tests **shall** be configured so that heat pipes are in gravity neutral or reflux orientations to ensure operation of these devices during thermal testing.

Reduced Thermal Math Models (CDRL SE-6b) of the instrument **shall** be delivered to the Government for use in designing the spacecraft-instrument interfaces. The detailed instrument models will be combined with the analytical models of the spacecraft to create a reduced system-level thermal model of the satellite. Only the instrument Reduced Thermal Model (RTM) will be delivered to the spacecraft contractor by the Government. The Government will deliver the spacecraft RTM to the instrument Contractor. Only the Government will have and run the integrated full models.

If there are any special requirements such as special temperature control, heater power, thermal distortions, etc., which are pertinent to the design, these considerations **shall** be identified and included in the analysis.

2.5.1.3 Reserved

2.5.1.4 Electromagnetic Compatibility/Electromagnetic Interference Analysis

The Contractor **shall** perform EMC/EMI analysis and/or test in accordance with the SMBA Performance Requirements Document (NS1-SMBA-REQ-0001) and MIL-STD-461 to verify the hardware meets the performance requirements of the equipment specification and provide an EMC/EMI Analysis Report documenting the analysis in accordance with CDRL SE-6d.

2.5.1.5 Radiation Hardness and Survivability Analyses

The Contractor **shall** perform BOL and EOL Hardness and Survivability analyses to verify the equipment meets the requirements in accordance with the PRD and MA-7 Master Electrical, Electronic, and Electromechanical (EEE) Parts List.

The radiation analysis report **shall** include radiation effects/degradations on components, coatings, and on electronic performance; Dose Mapping analysis; Single Event Effects Criticality Analysis; Displacement Damage analysis; parts total dose analysis; and test data supporting these analyses. Transient radiation effects that are not distinguishable from noise in the instrument output **shall** be included as a noise component in signal-to-noise ratio analyses and compliance evaluations. Transient radiation effects on temperature measurements used in any calibration **shall** be analyzed, including impacts on radiometric calibration.

2.5.1.6 Radiometric Math Model/Sensor Performance Math Model – Design and Development Documentation

The Contractor **shall** develop, deliver, and maintain a Radiometric Math Model/Sensor Performance Math Model (SPMM) in accordance with CDRL CV-4. It **shall** include assumptions and assessments of assumptions as to their fidelity. The SPMM is expected to supply and support system performance predictions.

2.5.1.7 Radio Frequency Interference Analysis

The Contractor **shall** iterate (approximately three iterations) RFI related requirements lessons learned through design, analysis, and test as necessary to develop the final requirements. The Contractor should work with the Government team to develop the design, conduct analysis, propose new requirement language, integrate the ETU, test, propose new requirement language, and iterate as necessary.

RFI analysis **shall** be in accordance with SE-6f.

2.5.1.8 Hyperspectral Capability Analysis

Hyperspectral Capability Analysis **shall** be in accordance with SE-6e.

2.5.1.9 Intercalibration Analysis

Intercalibration Analysis and Test Data **shall** be in accordance with CV-7.

2.5.2 Trending

The Contractor **shall** establish and implement a method for trending test data for all critical functions across missions through the life of the program in accordance with CDRL SE-2.

The trending method may differ between instrument level testing and satellite level. The Contractor **shall** coordinate with the Government the selected list of parameters to be trended in accordance with Trending Plan/Analysis Reports and Operations Log (CDRL SE-2). The Contractor **shall** monitor selected parameters for trends starting at the beginning of instrument electrical integration and continuing during the system integration and test phases through the on-orbit commissioning phase. The Contractor **shall** support trending of data across missions through the life of the program. The Contractor **shall** analyze the trended data for indications of anomalous conditions and for possible performance or reliability degradation. The Contractor **shall** provide Trending Reports and operating hours in accordance with the Trend Analysis, Reports, and Operations Log. The Contractor **shall** coordinate with the Government the list of parameters to be trended on-orbit. The Contractor **shall** support the development and validation of any Mission Operations Element operational trending system by participation in telecons as well as review and comment on documentation.

The Contractor **shall** review and report all trending results from every phase of the development with the Government.

2.6 Special Studies

The Contractor **shall** conduct, in addition to the requirements specified in this document and the Contract, additional engineering studies, tests, technical analyses, reviews of test results, design modifications, and tasks relating to the development, implementation, characterization, and operation of the instrument, as authorized by the Government and in accordance with Contract Section H Clause, Special Task Support Requirements. Each task will be initiated by written direction from the Government Contracting Officer. The Government will coordinate with the Contractor to define each task in detail, and establish manpower ceilings, performance schedules, and deliverables.

Special studies are performed under task orders issued against Contract Line Item Number (CLIN) 0003, defined in contract clauses GSFC 52.211-91, SCOPE OF WORK (FEB 2016), and 1852.217-72, PHASED ACQUISITION USING PROGRESSIVE COMPETITION DOWN-SELECTION PROCEDURES.

2.7 Constraints

The Contractor **shall** identify all constraints that pertain to all integration, test, and operational environments, including testing while encapsulated at the launch pad as a part of the ICD CDRL SE-8, including commands that can potentially damage the spacecraft or instrument, or cause injury to personnel.

3 MISSION ASSURANCE

The Contractor **shall** develop, implement, and maintain a comprehensive mission assurance program.

3.1 Equipment Tagging

All items (non-flight equipment) to be removed prior to flight **shall** be tagged with a Contractor's standard red tag, "(Remove Before Flight)."

All items to be installed at the launch site, such as instrument flight plugs, **shall** be considered "green tag" (Install Before Flight) items.

3.2 Mission Assurance Requirements

This section of the SOW establishes a risk-based application of mission assurance requirements (MAR) the SMBA instrument. It is organized around the SMA disciplines/focus areas listed in Table 3-1, each with its own section.

The requirements in this document use the following identification (ID) convention:
SMA Discipline Alphanumeric Identifier (e.g. QMS_01X).

Table 3-1. Requirements Identification Convention

Where the key is as follows: SMA Discipline	• GEN – General SMA Requirements • QMS – Quality Management System • SAF – System Safety • REL – Reliability • SWA – Software Assurance • WOR – Workmanship • EEE – EEE Parts • MAP – Materials and Processes • CON – Contamination Control • MAC – Metrology and Calibration • GID – GIDEP Alerts and Problem Advisories • EIA – End Item Acceptance Data Package
Numeric Identifier	A unique and sequential 2-digit number, with or without a character

3.3 General

3.3.1 Management

GEN_03a: The Contractor **shall** designate a manager for assurance activities.

GEN_03b: The assurance manager **shall** not be responsible for project costs and schedules other than those pertaining to assurance activities.

GEN_04: The Contractor **shall** ensure that the assurance manager has direct access to upper management that is independent of project management and with functional freedom and authority to interact with all elements of the project.

3.3.2 Suspension of Work Activities

GEN_11: The Contractor **shall** direct the suspension of any work activity that presents an unsafe work condition to personnel or imminent danger to property.

3.3.3 Surveillance

GEN_15: The contractor **shall** report the status of facility operations and quality metrics to NASA on a quarterly basis. This **should** include the following data for first tier and second tier suppliers of Project Level Critical Items.

- Quality escapes - Any product released by an internal or external supplier that is subsequently determined to be nonconforming to contract and/or product specification requirements.
- First pass yield - A measure of quality in a process that reflects the percentage of product made correctly without any rework or corrective activity.
- Supplier defect rate - The supplier defect rate measures the percentage of materials or product received from suppliers that do not meet required or compliance specifications.
- Internal audit results.

3.4 Quality Management System

3.4.1 General

QMS_01: The prime Contractor **shall** have a quality management system (QMS) that is third-party certified with the requirements of Society of Automotive Engineers (SAE) AS9100, "Quality Systems - Aerospace - Model for Quality Assurance in Design, Development, Production, Installation, and Servicing."

QMS_02: Lower-tier subcontractors and suppliers of procured critical hardware systems (e.g., components, subassemblies), piece parts, or suppliers using special processes determined to be critical (e.g., plating, polishing, soldering, brazing, additive manufacturing) **shall** maintain a current QMS that complies with one of the following: AS9100D (preferred), ISO9001:2015, AS9003B.

3.4.2 Material Review Board

QMS_06: The Contractor **shall** have a documented process(es) for the establishment and operation of an MRB to process major non-conformances.

3.4.3 Anomaly Reporting and Disposition

QMS_14: The Contractor **shall** have a documented process for the establishment and operation of an ARB to process (i.e., report and disposition) major anomalies, which are those that have resulted in hardware or software test failures and damage or potential damage to hardware.

QMS_16: Reporting of major anomalies **shall** begin with the first application of power at the component level, flight software acceptance testing, when interfacing with flight hardware, and the first mechanical operation.

3.5 Orbital Debris Assessment Report and End of Mission Plan

QMS_22: The Contractor **shall** provide the information necessary for the development of the Orbital Debris Assessment Report and the End of Mission Plan deliveries, per the content defined in NASA-STD-8719.14 (CDRL MA-2).

3.6 System Safety

3.6.1 General

SAF_01: The Contractor **shall** document and implement a system safety program and plan in accordance with NASA-STD-8719.24 with Annex, (System Safety Plan, CDRL MA-3).

3.6.2 Mission Related Safety Requirements Documentation

SAF_03a: The Contractor **shall** implement the launch range safety requirements that are applicable to the launch site. The sections below include the applicable requirements documents for the pertinent launch ranges.

3.6.3 System Safety Deliverables

(Reserved)

3.6.4 Safety Requirements Compliance Checklist

SAF_11: The Contractor **shall** document and implement a Safety Requirements Compliance Checklist, the Annex to NASA-STD-8719.24, to demonstrate that the payload complies with NASA and range safety requirements (Safety Requirements Compliance Checklist CDRL MA-4).

3.6.5 Instrument Safety Assessment Report

SAF_12: The Contractor **shall** generate an Instrument Safety Assessment Report (ISAR) to document the following: (ISAR, CDRL MA-5).

- Comprehensive evaluation of the risk being assumed prior to the testing or operation of an instrument.
- The Preliminary hazard analysis completed in accordance with NASA-STD8719.24 with Annex.
- The hazard verification tracking log completed in accordance with NASA-STD-8719.24 with Annex.
- Any hazardous procedures used during I&T or pre-launch processing at the launch site in accordance with NASA-STD-8719.24 with Annex.

The spacecraft contractor **will** use the ISAR (CDRL MA-5) as an input to the Safety Data Package.

3.7 Reliability

3.7.1 Reliability Program Plan

REL_01: The Contractor **shall** implement a Reliability Program that includes both qualitative and quantitative techniques to support decisions regarding mission success and safety throughout system development.

The Contractor **should** perform reliability analyses concurrent with design and maintain consistency between reliability analyses so that identified problem areas are addressed, and corrective action taken in a timely manner.

3.7.2 Analysis of Design

3.7.3 Failure Modes and Effects Criticality Analyses

REL_11a: The Contractor **shall** perform, document, maintain, and deliver a Failure Modes and Effects Criticality Analysis CDRL MA-6 in accordance with an industry Voluntary Consensus Standard (e.g., SAE) to identify and rank critical items that are being designed, built, or provided from project initiation through launch and mission operations.

3.7.4 Reliability Calculations

REL_21: The Contractor **shall** perform and maintain reliability and ground-system availability calculations to identify design weaknesses, support design trades, and demonstrate the impact of critical items.

REL_22: The Contractor **shall** perform a Limited Life Item Analysis (LLA) that includes expected life, required life, an assessment of life margin (including servicing and maintenance); provides retention rationale for items with an expected life of less than 2x the required life and fosters a plan to manage limited life items (Life Test Plan and Final Report for Lubricated Mechanisms CDRL MA-11).

The Contractor **shall** conduct a life test, within representative operational environment, to at least 2x expected life for all repetitive motion devices with a goal of completing 1x expected life by CDR.

3.8 Software Assurance

3.8.1 Software Assurance Program

SWA_03: The Contractor **shall** ensure the independence of software assurance from software engineering.

SWA_04: The contractor **shall** document and implement a Software Assurance Plan. NASA-STD-8739.8B is available as a guideline.

3.9 Workmanship

3.9.1 General

WOR_02: The Contractor **shall** implement a workmanship program to assure that electronic packaging technologies, processes, and workmanship meet mission objectives for quality and reliability per the requirements of the following standards:

- NASA-STD-8739.6 Implementation Requirements for NASA Workmanship Standards, excluding sections 8.1, 9.1, and 10.1
- IPC J-STD-001, Class 3
- GSFC-STD-6001
- IPC-2225
- IPC-6015

WOR_04: The Contractor **shall** comply with one of the following standards for electrical cables and harnesses:

- NASA-STD-8739.4
- IPC/WHMA-A-620x-S

3.9.2 Electrostatic Discharge Control

WOR_06: The Contractor **shall** implement an electrostatic discharge (ESD) control plan that conforms to the requirements of ANSI/ESD S20.20.

3.9.3 Printed Circuit Boards

WOR_08: The Contractor **shall** comply with one of the following standards for rigid printed circuit boards (PCB):

- IPC-6012 (Note: the latest revision preferred, but older revisions **may** be acceptable based on inherited designs or contractor standard practices)
- MIL-PRF-55110H
- ECSS-Q-ST-70-60

Note: In cases in which high-density interconnect components (e.g., reconfigurable field programmable gate arrays (FPGA), high-density static random-access memory, etc.) or board population are present, the Contractor **should** consult with the Project and Government Microelectronic Packaging and Circuit Board Commodity Risk Assessment Engineer for a specification strategy to avoid extensive rebuilds or the incorporation of risky features solely in order to meet the specification. This consultation **may** result in deviations from the specifications above with no waiver required.

WOR_10: The Contractor **shall** implement a PCB procurement plan.

3.9.4 Lead-Free Control Measures

WOR_16: Tin-based solders and surface finishes that are less than 3% lead by weight **shall** comply with the Level 2C requirements set from GEIA-STD-0005-1 and GEIA-STD-0005-2.

3.10 Electrical, Electronic, and Electromechanical Parts

3.10.1 General

EEE_02: The Contractor **shall** document and implement a Parts Control Plan per Level 3 instructions of GSFC EEE-INST-002. The Contractor should evaluate the use of Level 2 where appropriate to meet the mission life and radiation requirements.

The Contractor **shall** perform a parts stress analysis and make available upon request.

The Contractor **shall** develop and implement a Counterfeit Parts Control Plan.

The Contractor **shall** develop and implement an Obsolescence Parts Plan.

3.10.2 Parts Control Board

EEE_07: The Contractor **shall** establish a PCB that is responsible for the planning, management, and coordination of the selection, application, and procurement requirements of EEE parts.

3.10.3 Re-use of Electrical, Electronic, and Electromechanical Parts

EEE_15: The Contractor **shall** require approval of the PCB to re-use EEE parts that have been installed.

3.10.4 Master Electrical, Electronic, and Electromechanical Parts List

EEE_16: The Contractor **shall** develop and deliver a Master EEE Parts List and maintain it for the duration of the project (CDRL MA-7).

3.11 Materials and Processes

3.11.1 Materials and Processes Selection, Control, and Implementation Plan

MAP_01a The Contractor **shall** implement a Materials and Processes (M&P) Selection, Control, and Implementation Plan in accordance with NASA-STD-6016 (Standard Materials and Processes Requirements for Spacecraft) or own internal standard which includes an Additive Manufacturing Control Plan and Part Production Plan for the design and manufacture of additively manufactured parts in accordance with NASA-STD-6030 (Additive Manufacturing Requirements for Spaceflight Systems).

3.11.2 Materials Identification and Usage List

MAP_04 The Contractor **shall** prepare a Materials Identification and Usage List and maintain it for the duration of the project (MA-10).

3.11.3 Materials and Processes Compliance

The use of M&P that do not comply with the requirements of this NASA-STD-6016 may be acceptable in the actual hardware applications. Materials Usage Agreements are required for all M&P that are technically acceptable but do not meet the requirements of NASA-STD-6016, as implemented by the approved M&P Selection, Control, and Implementation Plan.

3.12 Metrology and Calibration

3.12.1 Metrology and Calibration Program

MAC_01: For measurement and test equipment that has documented requirements for metrological accuracy and traceability, the Contractor **shall** comply with NASA-STD-8739.12 or verify equipment against calibrated instruments or intrinsic standards, using a documented procedure.

Note: The Contractor **may** verify torque wrenches against a calibrated torque tester prior to use.

3.12.2 Use of Calibrated and Non-Calibrated Instruments

MAC_02: The Contractor **shall** record the measurements that require accuracy in applicable project build documents (e.g., work order authorizations, job orders, task sheets, or test plans), including the article of calibrated equipment used to take the measurement and its calibration end date.

MAC_03: When verification is chosen instead of calibration, the Contractor **shall** perform verification within a timeframe that has been demonstrated to provide appropriate levels of reliability, in the same facility, and under the same conditions that **will** be encountered during the process.

3.13 GIDEP Alerts and Problem Advisories

3.13.1 Government-Industry Data Exchange Program

GID_01: The Contractor **shall** participate in Government-Industry Data Exchange Program (GIDEP) per the GIDEP Operations Manual (Note: this document is available through <http://www.gidep.org>).

3.13.2 Alert Disposition

GID_03: The Contractor **shall** review the following, hereafter referred to collectively as Alerts, for effects on EEE parts, materials, equipment, and software used in NASA products:

- GIDEP Alerts
- GIDEP SAFE-ALERTS
- GIDEP Problem Advisories
- GIDEP Agency Action Notices
- NASA Advisories

GID_05: When the Contractor identifies an item in their design, inventory, or assembly that is documented in an Alert, the Contractor **shall** disposition the item and Alert through the MRB as a major nonconformance.

3.13.3 GIDEP Reporting

GID_06: The Contractor **shall** prepare and submit failure experience data and safety issue reports per the requirements of GIDEP Operations Manual whenever failed or nonconforming items are discovered that are available to other buyers.

3.13.4 Review Reporting

GID_07: The Contractor **shall** report the status of NASA products that are affected by Alerts (or by significant EEE parts, materials, software, and safety problems) and the actions taken to eliminate or mitigate negative effects at monthly status reviews, parts control board meetings, program milestone reviews, and readiness reviews.

3.14 End Item Acceptance Data Package

EIA_01: The Contractor **shall** maintain the EIADP throughout the project life cycle and submit in accordance with CDRL MA-13.

4 FABRICATION AND ASSEMBLY

The Contractor **shall** provide all necessary personnel, facilities, services, materials, tooling, and test equipment to fabricate and assemble the Instrument in accordance with its design specifications and Contract requirements. “Fabrication” in this sense also includes procured items.

A Manufacturing (Fabrication and Assembly) Flow Plan, or equivalent, **shall** be created and maintained and updates included as a part of reviews listed in Table 1-1, Top-Level Instrument Reviews.

4.1 Spares

The Contractor **shall** develop, deliver, and implement an approved spare parts program plan for critical flight spares and for critical GSE spares, including spares replenishment, built by the Contractor (e.g. Calibration Test Equipment, target assemblies, and electronics) in accordance with CDRL SE-12.

Once the plan is approved by the Government, the Contractor **shall** procure sufficient spare parts to account for typical attrition during the manufacturing process due to handling, random failure, programming, etc. The spare parts **shall** be qualified, tested, stored, and calibrated to the same standards as the flight instrument articles consistent with their spared level of assembly.

4.2 Software Development

The Contractor **shall** develop and deliver an Instrument Software Development Plan (SDP) based on the contractor’s established processes (Command Media) or a process based on NPR 7150.2.

For FPGA development, if utilized, reference the design guidelines, NASA GSFC 500-PG-8700.2.7, Procedures and Guidelines for Design of Space Flight Field Programmable Gate Arrays, and NASA GSFC 500-PG-8700.2.8, FPGA Development Methodology. If FPGAs are utilized the Contractor **shall** deliver CDRL SE-10 FPGA Design Data Package.

4.2.1 Software Development Plan

The Contractor **shall** develop and control the instrument flight software in accordance with the Instrument SDP (CDRL SW-1). The SDP **shall** apply to software developed for this Contract, whether developed by the Contractor or by a third party (i.e., commercial-off-the-shelf, Government-off-the-shelf, modified-off-the-shelf, subcontractor) and whether developed for flight use or for GSE/simulators.

The Contractor **shall** identify in the SDP software and heritage software reuse plans. The Contractor **shall** perform an analysis of unnecessary and/or unreachable code on the intended flight version of the instrument flight software and produce an Unreachable Code Analysis document to be made available for Government review. The analysis **shall** identify all instances (areas) of unnecessary/unreachable flight code, the general functionality associated with the code, the reason each is intended to be left within the flight load, and the justification (e.g., mitigating action) that explains why the included code does not pose a risk to the mission. The focus will be on technical risk to the long-term mission, not cost.

4.2.2 Software Requirements Specification

The Contractor **shall** submit an Instrument Software Requirements Specification (SRS) documenting the instrument flight software requirements (CDRL SE-5).

The Contractor **shall** perform all analyses and software systems engineering required to identify and allocate the software requirements (given the system and subsystem requirements). The Contractor **shall** ensure that all requirements are forward and backward traceable between system and software requirements and between software requirements, design, and test.

4.2.3 Software Detailed Design

The Contractor **shall** maintain software design details in accordance with Instrument Performance Specification CDRL SE-5 and **shall** deliver an Instrument Software Detailed Design Document (CDRL SW-2).

The Contractor **shall** document and maintain instrument GSE/STE software detailed design and make it available for Government review. Delivery of the GSE/STE Software Development Folders and Software Engineering Notebooks to the Government is not required, but they **shall** be made available for audit by the Government.

4.2.4 Instrument Software Development Environment

The Contractor **shall** develop the software development and validation environment for the instrument that supports the development and test of the flight software. It includes host development computer operating systems, high-level language compilers and debuggers, autocode generator software systems, machine language emulators, and test scenarios and procedures. It includes the software in the instrument test environment simulators that model the detectors and mechanisms. It also includes development support software such as configuration management systems for tracking software and document versions.

4.2.5 Software Testing

The Contractor **shall** plan and implement a software verification and validation program to ensure that software being developed or maintained satisfies functional and performance requirements defined in the SRS in accordance with the Instrument Integration, Test, and Verification Plan (CDRL IT-1).

The Contractor **shall** compile Software Test Descriptions, including detailed descriptions of all qualification and acceptance tests; Test Procedures (CDRL IT-3), including acceptance criteria for software prior to and after loading; and Test Reports/Results (CDRL IT-2), including the results of all qualification and acceptance tests.

4.2.6 Software Version Description

Instrument flight software and firmware code **shall** be delivered accompanied by a Software Version Description Document (CDRL SW-3) describing the deliverable code and the associated operating procedures. For all firmware media, Programmable Logic Devices and formatted and numbered drawings are required, with assembly numbers assigned to all loaded chips and drawing notes used to identify specific loading and test procedures, markings, etc.

4.2.7 Software User Manual

The Contractor **shall** prepare and deliver an Instrument User's Guide (CDRL OO-2) describing the installation, build, and operating procedures for the flight software.

4.2.8 Software Reviews

The Contractor **shall** report flight software status as part of the technical status portion of the Monthly Program Management Reviews (CDRL PM-1) whenever testing or changes to the flight software baseline are being planned or are occurring.

The Contractor **shall** prepare for and conduct a Flight software acceptance TRR and deliver a Software TRR Data Package (CDRL RE-13).

The Government **shall** be invited to attend all Qualification TRR and Formal Qualification Testing.

4.2.9 Software Measures (Metrics)

The Contractor **shall** acquire software measures (metrics), as defined in the SDP CDRL SW-1, from any subcontractors or team members and provide them to the Government both as raw data and in graphical form as part of the Monthly Program Management Review (see CDRL PM-1). The collected software measures will support the analysis of both software product quality and schedule/effort/cost performance. The collection and reporting of metrics will be automated to the fullest extent practical.

4.2.10 Operational Simulator Software

The Contractor **shall** develop and maintain Simulator Software detailed design.

4.2.10.1 Test and Verification

The Contractor **shall** plan and implement the testing of instrument-spacecraft interfaces with the Spacecraft-Instrument Interface Simulator (SIIS) provided by the spacecraft contractor (**TBR**). Technical support will be provided by the spacecraft contractor at the instrument simulator provider facility. This will support early risk reduction of instrument-spacecraft interfaces.

The instrument Contractor **shall** support the spacecraft activity to verify the performance of the flight software by providing flight quality procedures and scripts and providing technical support to the spacecraft contractor performed testing including completing a Government approved Day-In-The-Life test which will be conducted at the Spacecraft simulator provider facility.

4.2.10.2 Simulator User's Guide

The Contractor **shall** update and deliver the Instrument Simulator User's Guide (CDRL OO-6) describing the installation, build, and operating procedures for the Simulator Software.

4.2.11 NASA Independent Verification and Validation Support

The Contractor **shall** ensure that all software documentation and code required for NASA's Software IV&V effort is made available to NASA IV&V personnel if directed by the Government. This includes access to all software reviews and reports, developer plans and procedures, software code, software design documentation, and software problem reporting data. Wherever possible, the Contractor **shall** permit electronic access to the required information or

furnish soft copies of requested information to NASA IV&V personnel.

The Contractor **shall** review and disposition all NEON Program-approved IV&V findings and recommendations. A Contractor point of contact **shall** be assigned and available to NASA IV&V personnel, as required, for questions, clarification, and status meetings.

4.2.12 Software Maintenance

The Contractor **shall** prepare and deliver an Instrument Software Maintenance Plan (CDRL SW-4) describing the plans, products, schedule, tasks, resources (software, hardware, facility) requirements, risks, and recommended contingencies for the maintenance of the flight software, instrument simulators, and ETU/GSE/STE systems.

The Contractor **shall** develop and maintain the Instrument flight software, simulators, and documentation, along with the environments, emulators, and test software necessary to develop and verify these systems through the duration of the Contract.

Any Contractor developed software development and/or validation tools (test bed) used by the Contractor to qualify and maintain the Instrument flight software **shall** be maintained by the Contractor for the duration of the Contract.

The Contractor **shall** maintain any special hardware (non-commercial such as command and data handling (C&DH) components and power supplies) for the duration of the Contract.

The delivery of a flight software patch or update **shall** be executed by sending a CD with the patch or uploading the patch to a Government SharePoint folder.

4.2.13 Sensor Data Sets

The Contractor **shall** prepare and deliver sensor test data sets simulating 72-hour continuous science data (CDRL SW-5).

4.3 Hardware

The Contractor **shall** deliver hardware as defined in the Contract.

4.3.1 Engineering Test Unit

The NASA Systems Engineering Handbook NASA SP-2016-6105 defines an Engineering Test Unit as: “A high-fidelity unit that demonstrates critical aspects of the engineering processes involved in the development of the operational unit. ETUs are intended to closely resemble the final product (hardware/software) to the maximum extent possible and are built and tested so as to establish confidence that the design will function in the expected environments.” A high-fidelity ETU is a critical tool for the program to manage technical and programmatic risk.

The Contractor **shall** conduct an Engineering Development Program to raise confidence in the instrument design and to minimize risk, cost, and schedule for the FM development, this will include the EPRs called out in Section 2.4, Design Engineering. At a minimum, the Engineering Development Program **shall** include development of a stand-alone ETU that is representative of the form, fit and function of the FMs.

The Contractor **shall** utilize a serial number to identify the ETU. The ETU **shall** be built from formally released drawings, utilizing Contractor's approved parts, materials, processes, and procedures. The ETU **shall** be successfully tested by the Contractor in accordance with the Specification and the approved test procedures/levels/durations (CDRL IT-3). Test procedures may be tailored as necessary to achieve the equivalent of acceptance as defined in the ICD and PRD. Any proposed tailoring **shall** be in accordance with CDRL IT-1.

The ETU should be used to:

- Evaluate the flight hardware design.
- Evaluate flight software.
- Evaluate FM test program.
- Qualify the instrument design.
- Dry-run the assembly build process, develop assembly and test procedures, initial run of test procedures, and test scripts, and identify any necessary mechanical GSE.
- Validate electrical and data interfaces design.
- Long term use for reach back and reach forward investigative support for anomalies, and updates for the flight instrument series.
- Identify and incorporate design updates for test procedure, execution improvement updates, and GSE updates.
- Develop and evaluate constraints.
- Validate thermal models.

The ETU **shall** be a flight qualified.

Prior to delivery of the ETU the unit **shall** be environmentally qualified and Calibration/Validation (Cal/Val) complete.

The ETU **shall** be vacuum compatible, mechanically equivalent, and electrically equivalent, and use the same fabrication methods as the FMs to the extent possible. The ETU may be used for the development of Recommended Operating Procedures (ROP), validation of GSE, etc.

The Contractor **shall** prepare and deliver an ETU Development and Test Plan in accordance with CDRL IT-1 and submit an ETU Requirements Alignment Matrix that documents differences between the ETU and the flight instrument at PDR and CDR, in accordance with CDRL RE-4 and RE-5. These CDRLs will serve as the backbones for the ETU Pre-Build Review and ETU Test Summary Review.

The build, assembly, and ambient functional test of the ETU **shall** be completed prior to the Instrument-level CDR. If there is a conflict between the ETU Ambient Functional Testing and the CDR that will drive the critical path schedule, the Contractor **shall** notify the Government and attain Government approval if the CDR is held prior to the ETU ambient functional testing completion.

The Contractor **shall** retain and maintain the ETU at the Instrument Contractor's facility during the FM development, test, and operations for the duration of the program.

4.3.2 Flight Instruments

The Contractor **shall** utilize a serial number to identify each instrument. The flight units **shall** be built from formally released drawings, utilizing Contractor's approved parts, materials, processes, and procedures. The flight units **shall** be successfully tested by the Contractor in accordance with the Specification and the approved test procedures/levels/durations (CDRL IT-3). Test procedures may be tailored as necessary to achieve the equivalent of acceptance as defined in the ICD and PRD. Any proposed tailoring **shall** be in accordance with CDRL IT-1.

The Contractor **shall** provide the survival heaters that will be powered by the spacecraft.

4.3.2.1 First Flight Model

The Contractor **shall** develop and deliver the first SMBA FM, FM1. FM1 **shall** be tested to protoflight levels as per Section 7 of the PRD.

The FM1 effort **shall** include fabrication, assembly, test, maintenance, storage, delivery, post-delivery, post-launch, and on-orbit operations support per the Contract.

4.3.2.2 Additional Flight Models

The Contractor **shall** develop and deliver additional FMs as per the Contract.

The additional FMs **shall** be tested to acceptance levels as per Section 7 of the PRD. If modifications to the design are made between FMs, the Contractor and the Government will come to agreement on the test levels.

Note: For TVAC verification, all hardware, including follow-on units, should be tested to protoflight levels per the PRD.

The effort **shall** include fabrication, assembly, test, maintenance, storage, delivery and post-delivery, post-launch, and on-orbit operations support as per the Contract.

4.3.2.3 Spares

The Contractor **shall** procure any necessary additional spares to ensure continued compliance with the requirements in Section 4.1.

The Spares Plan **shall** be updated in accordance with CDRL SE-12.

4.3.3 Ground Support Equipment

The Contractor **shall** develop and maintain GSE, including the STE, in accordance with the Contract. The GSE **shall** include specialized equipment necessary to support and accommodate the instruments at all facilities during I&T, transportation, storage, satellite I&T, launch site, and any other location. This should include GSE spares.

The Contractor **shall** plan for and provide the following information regarding GSE in IT-1 and IT-4:

- a) Number and dimensions of racks.
- b) Location of racks.
- c) Phase of I&T the test racks are required.

d) Power requirements.

The Contractor-provided Mechanical/Electrical GSE **shall** include all items necessary for testing, deploying, stowing, and protecting the instrument, including, but not limited to:

- STE – Interfaces to the SMBA instrument to:
 - Provide power.
 - Monitor health and status.
 - Record and store data/files.
 - Simulate interfaces to the spacecraft (e.g., C&DH, etc.).
 - Control and monitor calibration targets.
 - Instrument commanding.
 - Provide timing.
- Hard Protective Cover for antenna apertures.
- Mechanical GSE for use during Vibration testing and Thermal Balance/TVAC testing.
- Drill template.
- Electrostatic discharge bags.
- EMI cover.
- Calibration System for Ambient Testing.
- Calibration Targets (for TVAC Instrument Level Testing).
- RF blocker for EMI Testing.
- Spacecraft Emulator (delivered from the spacecraft provider).
- Flight software and Firmware Test Bed.
- Spacecraft test and operations language GSE.
- RFI generator.

External GSE Calibration Targets configuration **shall** be capable of accommodating Government provided targets in place of Contractor targets.

The Contractor **shall** provide two (TBR) additional sets of thermal control equipment for exclusive use at the spacecraft integration facility(ies).

4.3.4 Operations Simulator

The SMBA Operations Simulator (OpSim) is used by the NEON Series-1 team to prepare the launch readiness activities, verify satellite operational loads, flight software patches, maneuver planning, ground system development, Verification and Validation (V&V) and maintenance of operational procedures and testing, flight product validation, test/operations procedure development and validation, assist in anomaly resolution, and operator training.

The Contractor **shall** develop, test, and deliver three (3) OpSim that provides the following:

- Consists of flight like electromagnetic board and flight software.
- Provides realistic sensor operation and responses; including mission data.
- Provides capability to inject real-time or scripted anomalies.

The SMBA OpSim functionality **shall** consist of the following:

- Executes the flight software binary image unchanged.
- Runs real-time and advanced time.

-
- Ability to suspend and resume scenario-based testing.
 - Supports fault injection.
 - Simulated Solid State Recorder – same operations as user interface.
 - Used by operations to verify software and data loads to the vehicle and support operations maintenance.
 - Operator Training.
 - ROP/Procedure (PROC) development.
 - Interface checkout (e.g. Ethernet, 1pps, timing, etc.)
 - Commanding
 - Data
 - Telemetry

In order to support the development of scripts for use at the satellite level the OpSim **shall** be delivered prior to the instrument PER.

To assist in the development, testing, and operational evaluation of flight software patches, the Spacecraft Simulator **shall** run the same flight software image(s) as the actual flight system.

The OpSim **shall** operate in real-time.

NOTE: In this context, “real-time” means that one unit of wall-clock time equals one unit of simulation time, and that all interface timing requirements for the actual flight software executing in the environment/context of its host flight hardware platform will be simulated in like manner.

4.3.5 Software Testbench

The Contractor **shall** develop and test a SMBA Software Testbench with the following functionality:

- a) Command and data handling.
- b) Scheduling of events.
- c) Interrupt handling.
- d) Control of interfaces.
- e) RFI detection algorithms.

4.4 Deliverable Documentation

Deliverable documentation is specified in the CDRL. All data **shall** be delivered to the Government as specified in the Contract. Electronic submittal of CDRL items is required. Delivery dates shown in the CDRL are the dates the documents are to be received at the Government’s facility.

Each data item **shall** be prepared in accordance with the requirements defined in this SOW and accompanying CDRL. The Contractor may supply existing documentation if the existing documentation encompasses the requirements defined in this SOW. All CDRL items **shall** be delivered for Approval, Review, Information, or as Previously Delivered Acceptance. See the CDRL for the definition of each category.

For all document submissions, the Contractor **shall** send an email with a formal contracts letter to the Government informing the Government of the virtual delivery of the document.

CDRL submissions **shall** be complete, unabridged, and unredacted. To the extent that Proprietary and controlled unclassified information data is employed to support program execution, such data **shall** be appropriately marked and included.

The Contractor **shall** be responsible for the packaging and delivery of documents and data in the CDRL.

All CDRL submittals **shall** be protected by encryption or through the use of other secure means approved by the Government against unauthorized disclosure including interception and decryption by foreign nationals. Prior to delivery, the Contractor **shall** implement all safeguards necessary to ensure data is protected in accordance with the Export Administration Regulations except when expressly authorized by a Technical Assistance Agreement.

5 TEST PROGRAM

The Contractor **shall** prepare and conduct a test program that meets qualification, protoflight, and acceptance levels using GSFC-STD-7000, General Environmental Verification Standard (GEVS), as a guideline and as specified in the SMBA PRD. The test program **shall** demonstrate that all applicable specified requirements are satisfied. The test program **shall** include tests as necessary at all levels of assembly. The hardware test program **shall** cover all phases of the development and manufacture in accordance with the requirements identified in this SOW and the PRD. The Contractor **shall** provide a plan that includes any recommended tailoring for the instrument environmental tests for Government approval in accordance with CDRL item IT-1.

All composite structures and structural bonded joints **shall** be proof tested, regardless of safety factor.

Prior to mating with other hardware, electrical harnessing **shall** be tested to verify the wire routing, isolation, impedance, and overall workmanship. The following parameters **shall** be verified as minimum as documented in the ICD:

- a) Accuracy (signals on correct pins and nowhere else).
- b) Inputs and outputs (unloaded and loaded).
- c) Specified range (high/low extremes as well as nominal).
- d) Range impacts (how range extremes of one signal affect related signals).

Cabling electrically and physically equivalent to flight cables **shall** be used for all EMC testing. A Government agreed-to design, constructed with expected flight wire groupings, twisting, shielding, and shield terminations. The Contractor **shall** conduct a test program to demonstrate that all applicable specified requirements are satisfied. The test program **shall** include tests as necessary at all levels of assembly.

The Contractor **shall** provide access to facilities and data to support independent review of test data by the Government team during instrument performance and environmental tests. Tools utilized by the Contractor to verify instrument performance **shall** be made available to the Government.

Contractor **shall** take and maintain test data in accordance with the Sensor Performance Specification, the approved test plan (CDRL IT-1) and the requirements. Problems, anomalies, failures, etc., **shall** be reported and treated in accordance with the MA section of this SOW for investigation, resolution, and corrective action.

Comprehensive Performance Tests (CPT) **shall** be conducted at the subsystem, instrument, and satellite levels of assembly to demonstrate that the hardware and software meet their performance requirements within allowable tolerances.

CPTs **shall** demonstrate:

- a) With the application of known stimuli and appropriate inputs, the test article will produce the expected responses and outputs within allowable tolerances.
- b) The operation of redundant circuitry.
- c) Satisfactory performance in selected operational modes.

- d) When compared to the initial baseline CPT, performance of the test article has been unaffected by the environmental testing program.

When environmental testing is performed at a given level of assembly, CPTs **shall** be conducted during the hot and cold extremes of the test for both maximum and minimum input voltages.

Following integration of the instruments, a CPT will be executed in the configuration expected for environmental testing. Testing **shall** include operational performance evaluation at minimum and maximum flight voltages.

Limited Performance Tests (LPT) **shall** be conducted at the subsystem, instrument, and satellite levels of assembly, where CPTs are not warranted, to demonstrate that the hardware and software performance has not been degraded by environmental testing/exposure. LPTs are a specific subset of the overall CPT test sequence.

LPTs **shall** demonstrate:

- a) With the application of known stimuli and appropriate inputs, the test article will produce the expected responses and outputs within allowable tolerances.
- b) The operation of redundant circuitry.
- c) Satisfactory performance in selected operational modes.
- d) When compared to the initial baseline LPT, performance of the test article has been unaffected by the environmental testing program.

Electrical Aliveness Tests **shall** be conducted at the satellite level of assembly, where LPTs and CPTs are not warranted, to verify nominal telemetry response after power on and to demonstrate, to the extent possible by the test configuration, performance has not been degraded by environmental testing/exposure. Electrical Aliveness Tests are a specific subset of the overall CPT/LPT test sequence.

The instrument **shall** be capable of operating in ambient conditions without interruption of testing at the satellite level. This may require thermal control GSE to maintain instrument temperature during ambient testing. For satellite-level aliveness tests, the instrument **shall** complete aliveness testing without constraining satellite operations and without requiring thermal control GSE.

At the satellite level, Ground Compatibility Tests (GCT) **shall** be performed to demonstrate the Mission Operations Center's ground system capability to communicate, command, and control the instrument (up-link and down-link) via satellite mission communications links.

GCTs **shall** demonstrate:

- a) Simulated normal orbital mission scenarios encompassing launch, systems turn-on, housekeeping, command/control, and stabilization/pointing, including the collecting, processing, and archiving of science data.
- b) That the instrument is immune to erroneous commands.
- c) Autonomous Safe Mode and simulated anomaly recovery operations.

Items that incorporate changes in design, manufacturing processing, environmental levels, or performance requirements **shall** be re-qualified.

Qualification by similarity **shall** be permitted only with the concurrence of the Government.

The following is a sample SMBA Instrument Test Flow:

- Spectral Response Function (SRF) Test
- Antenna Pattern Measurement Test
- First CPT
- Antenna Alignment Test
- Vibration Tests
- Antenna Alignment Test
- Momentum Compensation Test
- Weight and Center of Gravity Test
- EMI Tests
- RFI Detection Algorithm Testing
- TVAC Cycling Test
- TVAC Balance Test
- TVAC Calibration Tests
- Antenna Alignment Test
- Final CPT

SRF and antenna pattern measurements **shall** be measured as defined below. All other tests in the Suggested Test Flow **shall** be done in accordance with the requirements specified in the SMBA PRD.

5.1 Spectral Response Functions

The SRF **shall** be measured and fully documented for each band/channel of each instrument in accordance with PRD requirements in PRD Section 4.10 and CDRL CV-6.

SRF data **shall** be characterized at a minimum of three temperature conditions: coldest predicted operational temperature, warmest predicted operational temperature, and the median temperature between the coldest and warmest predicted operational temperatures.

SRF data **shall** accommodate SRF changes over the operational lifetime of the instrument.

The gain range of SRF data **shall** be at minimum the achieved channel rejection magnitude per the PRD and measure the noise floor. SRFs need to accurately represent the noise floor.

5.2 Radio Frequency Interference

The Contractor **shall** test the SMBA instrument's capability to detect man-made RFI in the bands identified in PRD Table 4.2.1.

The Contractor will decide the best means of doing this (e.g., data sets, data ports, RFI generator, RFI Test Bench, etc.).

RFI testing **shall** include an end-to-end test.

The Contractor **shall** develop and deliver RFI test cases and perform RFI detection analysis (CDRL SE-6f) for what is currently seen from both terrestrial and space-based sources and from what can be reasonably projected in the future.

The algorithms utilized in the detection schemes **shall** be chosen by the Contractor and approved by the Government.

5.3 Antenna Pattern Measurements

The antenna radiation characteristics of each instrument ETU and FM **shall** be measured and documented by the Contractor. For each SMBA channel, measurements **shall** include beam pointing, beamwidth, beamwidth variation, and beam efficiency.

The Contractor **shall** permanently affix an optical alignment cube that **shall** be used to reference antenna pattern measurements and provide the Spacecraft integrator a known relative indicator of boresight alignment in accordance with PRD Section 5.7.1.

For SMBA analog frequency bands, antenna patterns **shall** be measured at the center frequency of the corresponding frequency band.

For SMBA hyperspectral frequency bands, antenna patterns **shall** cover the frequency bands as defined in the PRD in Section 4.1. At a minimum, antenna patterns **shall** be measured at the two band edge frequencies and at the band center frequency. For V-Band, an additional two frequencies (one on each side midway between the band edge and the band center) **shall** be measured in accordance with CDRL SE-6e.

5.3.1 Plane Pattern Cuts

At each frequency, a minimum of three cuts **shall** be made to obtain radiation patterns.

These cuts **shall** be the principal plane of the antenna (plane of symmetry of the antenna, also referred to as the down track plane), a plane 90° from the principal plane (also referred to as the cross track plane), and a 45° diagonal cut.

5.3.2 Measurement Polarization

For all channels, each cut **shall** be measured at two polarizations, Co-polarized (0°) and Cross-polarized (90°). Note: The 0° is parallel to the principal direction of polarization at the electrical boresight (beam center) of the beam position in question.

5.3.3 Dynamic Range

The minimum dynamic range of the antenna pattern measurements **shall** be 70 dB.

5.3.4 Amplitude Accuracy

The amplitude accuracy of the antenna pattern measurements **shall** be less than +/- 3% in dB values throughout the dynamic range.

5.3.5 Angular Range and Interval

Each pattern **shall** cover a total angular range of 360°, i.e., plus and minus 180° from electrical boresight. The angular intervals **shall** be a maximum of one-fifteenth (1/15th) of the antenna Half Power Beamwidth.

5.3.6 Beam Positions Required

The antenna pattern measurements **shall** be made at beam positions corresponding with:

- a. Beginning, middle, and end of the Earth-view sector.
- b. Cold calibration reference beam position.
- c. Warm calibration reference beam position.

5.4 Test Readiness Reviews

TRRs for the instrument are defined in Section 1.5.1.9.

TRRs for instrument subsystems **shall** be performed as part of the applicable subsystem peer reviews, as defined in the Engineering Peer Review Plan (CDRL PM-2).

5.4.1 Test Process Decision Meetings

The purpose of Consent to Proceed (CTP) and Consent to Break (CTB) are to ensure test objectives have been met, review quicklook test and science data, and to review any anomalies prior to breaking configuration. The Contractor **shall** conduct, document, and invite the Government to CTPs and CTBs (CDRL IT-3) as informal EPRs and make available upon request any data presented or referenced in support of any decision related to the forward processing of the Instrument. CTPs/CTBs **shall** be conducted at the end of the initial CPT, EMI/EMC, Vibration, shock and acoustic, TVAC, at plateau changes in TVAC, and at final CPT. The Contractor **shall** establish criteria ahead of the meetings and evaluate whether the criteria were met.

5.5 Pre-Environmental Review

The Contractor **shall** present the description and results of the Instrument Pre-Environmental Test program and demonstrate their readiness for environmental testing. The PER Data Package (CDRL RE-6) **shall** contain all relevant instrument information required to satisfy paragraph 9.2 (Criteria for a Successful System Integration Review) of GSFC-STD-1001, Criteria for Flight and Flight Support Systems Life Cycle Reviews.

5.6 Instrument Test and Calibration

5.6.1 Instrument Program Test Plan

The Contractor **shall** prepare and deliver a test plan describing the methods, policies, organization, and flight unit test program in accordance with CDRL IT-1. The Contractor may submit separate plans for each phase of the test program or one plan covering all phases.

5.6.2 Instrument Calibration

The Contractor **shall** perform through the antenna thermal vacuum calibration of the SMBA ETU and all FMs.

The system calibration test program **shall** produce the information required to evaluate the orbital measurement with a noise equivalent differential temperature (NEdT) precision better than the required values listed in the PRD.

The system TVAC calibration test program **shall** emulate “test-as-you-fly” conditions.

The Contractor **shall** collect calibration data in instrument operational mode.

The calibration environment for the instrument **shall** emulate the mission thermal radiation environment.

The Contractor **shall** provide a variable Earth scene target with a radiometric brightness temperature that varies from 80K to 330K.

The Earth Scene Target temperature **shall** be capable of being set to within ± 1 K of its pre-selected value. The target temperature is determined from the weighted mean of temperature sensors imbedded in it.

The Contractor **shall** provide a cold space calibration target representative with a radiometric brightness temperature as close to the on-orbit cold space brightness temperature as possible. No warmer than 80K is desired.

The target temperatures are determined from the weighted mean of temperature sensors imbedded in them. During the period that calibration data is taken, the indicated temperature change of each sensor taken individually **shall** be less than 0.1 K. If any sensor indicates a target temperature change in excess of 0.1 K, all data for that run will be discarded.

The Contractor **shall** perform calibration at a minimum of three instrument temperatures representing the SMBA instrument low operational, mid operational, and high operational temperatures (TBS). Actual instrument operational temperatures and control temperature locations will be agreed upon between the Contractor and the Government.

The instrument temperature **shall** be set to within ± 1 K of its preselected value. The temperature will be stabilized during the data run for any given target temperature. If the interface temperature varies by more than ± 1 K, all data for that run will be discarded.

The instrument internal temperature **shall** be taken at one point within the instrument where there is a flight temperature sensor. The temperature **shall** be set within ± 1 K and stabilized sufficiently during the data run for any given target temperature. If the instrument internal temperature varies by more than ± 0.5 K, all data for that run will be discarded.

The Contractor **shall** perform hysteresis measurements at a minimum of two variable Earth scene target temperatures at each of the instrument operational temperatures. Actual target temperatures will be agreed upon between the Contractor and the Government.

The Contractor **shall** prepare and deliver a Calibration Plan that includes all aspects of calibration both ground and on-orbit. This will include intercomparison and intercalibration as part of a Cal/Val and Intercalibration Plan per CDRL CV-1. GSFC owns the primary standard. Intercalibration will be accomplished via National Institute of Standards (NIST)-traceable transfer standards. GSFC may create NIST-traceable transfer standards. Intercalibration of the SMBA instrument **shall** be performed using Government furnished equipment transfer standards or the instrument provider's transfer standard (**TBR**).

Procedures for Thermal Vacuum Cal/Val **shall** be provided in accordance with the Cal/Val Procedures CDRL CV-2. Procedures for Cal/Val and intercalibration **shall** be provided in accordance with CDRL CV-2.

Cal/Val results and summaries for the ETU and each FM **shall** be provided in accordance with Cal/Val Results CDRL CV-3.

A Radiometric Math Model for the SMBA Instrument **shall** be developed and provided in accordance with the Radiometric Math Model CDRL CV-4.

A Calibration Data Book for the ETU and each FM **shall** be provided in accordance with CDRL CV-5.

Pre-launch calibration parameters required for data processing **shall** be delivered in accordance with CDRL CV-5.

Calibration, intercalibration, and data processing algorithms **shall** be provided in accordance with CDRL CV-7.

The Contractor **shall** perform Point & Stare operations as defined in PRD Section 4.8.4, Scan Characteristics, and Section 5.6.2.4.2, Instrument Diagnostic Mode, at a minimum of two variable Earth scene target temperatures at each of the tested instrument temperatures to determine the Noise Power Spectral Density (NPSD) curves like that shown in Figure 5-1 for each SMBA channel as defined in PRD Table 4.1.2. The NPSD is an important characteristic of all radiometer channels relating quantities like NEdT, 1/f noise, gain variations, and striping. Actual target temperatures will be agreed upon between the Contractor and the Government.

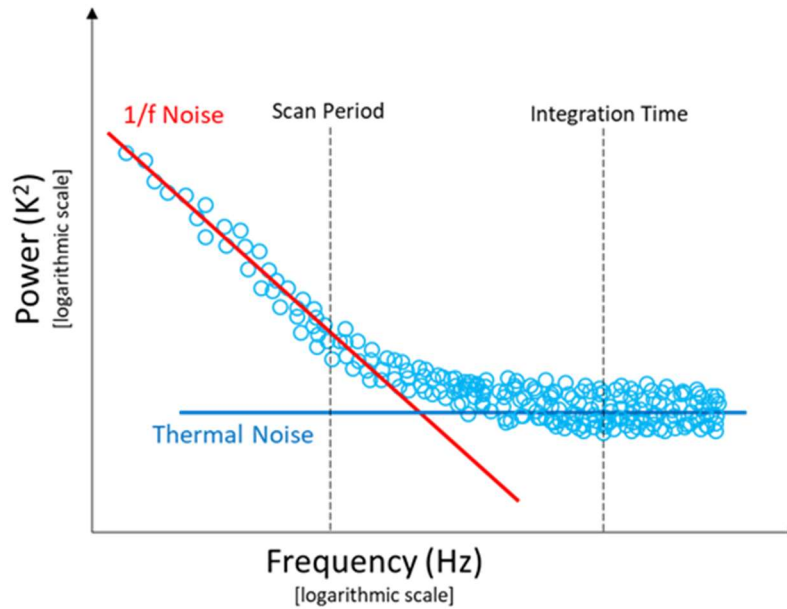


Figure 5-1. Notional Noise Power Spectral Density

The Contractor **shall** prepare and deliver a Cal/Val and Intercalibration Plan in accordance with CDRL CV-1 describing tests to be conducted for calibrating the radiometric response of the instruments against each other and a ground-based standard.

The Contractor **shall** prepare and deliver the SRF data for the SMBA Instrument in accordance with the SRF Measurements CDRL CV-6.

5.6.3 Calibration Test Equipment

The Contractor **shall** provide one set of Calibration Test Equipment (CTE).

The CTE **shall** be designed and built to be used in a thermal vacuum environment.

The CTE **shall** include calibration reference targets representing the on-orbit cold space view scene and on-orbit Earth viewing scene.

All CTE targets **shall** have a target emissivity of no less than 0.9999 for all SMBA channels.

All CTE targets **shall** be large enough to fill the instrument field of view when the instrument is pointed toward the target.

The CTE cold space view calibration reference target(s) **shall**:

- Contain 4-wire platinum resistance thermometers (PRT) imbedded and located in the target sufficient to measure the physical temperature of the target and measure the vertical and horizontal temperature gradients within the target.

The CTE Earth viewing scene calibration reference target(s) **shall**:

- Produce a minimum of 11 radiometric brightness scene temperatures corresponding to target physical temperatures from 80K to 330K in 25K steps.
- Contain a number of 4-wire PRTs imbedded and located in the target sufficient to measure the physical temperature of the target and measure the vertical and horizontal temperature gradients within the target.

5.6.4 Test Software Validation Plan

If automated or semi-automated testing is planned or used, the Contractor **shall** carry out a software validation process as documented in the Instrument SDP (SOW Section 4.2.1).

5.6.5 Test Procedures

The Contractor **shall** prepare Detailed Test Procedures and As-Run Test Procedures for all hardware to be delivered in accordance with CDRL IT-3. For those supporting documents that contain items that are unique to specific units (i.e., test data and configuration information), representative documents **shall** be submitted.

The Contractor **shall** deliver as-run, instrument test procedures in accordance with CDRL IT-3. The Government will be responsible for any translation of test procedures to ensure compatibility with Government hardware or software.

5.6.5.1 Notification of Tests

The Contractor **shall** notify the Government no later than five days in advance of instrument and lower-level acceptance tests. Tests will not be delayed if Government's representative is not available for any test unless Government presence is required per the Contract, i.e., mission assurance requirements.

5.6.6 Test Reports and Test Data

Test results **shall** be made available after completion of each test in accordance with CDRL IT-2.

5.7 Verification and Validation

5.7.1 Verification & Validation Plan

The Contractor **shall** develop and deliver a comprehensive V&V Plan and associated Requirements Verification Matrix in accordance with the System Performance Verification Plan and Requirement Verification Matrix CDRL SE-4.

The following delineates the baseline verification testing sequence for the SMBA instrument and, while cannot be deleted, the sequence of tests can be tailored as needed to optimize schedule or workflow considerations.

The baseline instrument verification tests, prior to delivery of the instrument, **shall** be the following:

- a) Electrical Interface Testing
 - Including harness testing
- b) CPT
 - Including LPTs
 - Including Mechanical Deployments
- c) EMI/EMC Characterization and Verification tests

- a. Including LPTs between EMI/EMC test types
- d) LPT
- e) Structural and Dynamic Testing
 - Pre-Test Mechanical and Antenna Alignment Verifications
 - Vibroacoustic Testing
 - Sinusoidal Vibration Testing
 - Shock Testing
 - Post-Test Mechanical and Antenna Alignment Verifications

Note: LPTs should be performed between each Structural Test sequence (each axis or sequence where appropriate)
- f) CPT (pre-TVAC, and prior to each environmental test)
- g) TVAC and Thermal Balance Testing
 - Including Plateau CPT and LPT Testing
 - Including Performance Verification Testing
 - Including Contamination/Bake-out Testing
- h) CPT (post-TVAC, and after each environmental test)
 - a. Including Final Mechanical Alignment Verifications
- i) Mass Properties

The SMBA instrument **shall** survive the Satellite-Level Test program. The baseline instrument verification tests, **after shipment to the Satellite integrator**, are defined as:

- a) Electrical Interface Tests
 - Including Electrical Aliveness testing
- b) CPT
 - Including Mechanical Deployments
- c) EMI/EMC Characterization and Verification tests
- d) LPT
- e) Structural and Dynamic Testing
 - Pre-Test Mechanical Alignment Verifications
 - Vibroacoustic Testing
 - Sinusoidal Vibration Testing
 - Shock Testing
 - Post-Test Mechanical Alignment Verifications

Note: LPTs should be performed between each Structural Test sequence (each axis or sequence) where appropriate.
- f) CPT (pre-spacecraft TVAC, and prior to each environmental test)
- g) TVAC and Thermal Balance Testing
 - Including Plateau CPT and LPT Testing
 - Including Performance Verification Testing
 - Including Contamination/Bake-out Testing
- h) CPT (post-spacecraft TVAC, and after each environmental test)
 - Including Mechanical Deployments
 - Including Final Mechanical Alignment Verifications
- i) Mass Properties
- j) GCT

The Contractor will analyze and document results for those tests recommended by the Contractor that are run at Satellite levels to demonstrate that the instrument hardware and software meet their performance requirements within allowable tolerances.

Ground testing of the integrated instrument flight software with the spacecraft ground system **shall** accumulate a minimum of 72 hours of error-free, uninterrupted testing using highest fidelity flight software test bed.

5.7.2 Verification Reports

The Contractor **shall** provide verification/test reports in accordance with CDRL IT-2.

6 DELIVERY

The Contractor **shall** be fully responsible and liable for the preservation, packaging, product marking, packing and delivery of all hardware and documentation in accordance with this SOW and all applicable documents of Sections VI and VII to assure safe arrival at destination.

6.1 Acceptance or End Item Acceptance Data Package

An acceptance or EIADP **shall** be prepared for each flight deliverable instrument and the ETU. The requirements for the EIADP are specified in the CDRL MA-13 data item description.

6.2 Packaging, Handling, Storage, and Transportation

The Contractor **shall** provide Packaging, Handling, Storage, and Transportation (PHS&T) Plans and Procedures in accordance with the Instrument Post Delivery Support (PDS) Document CDRL IT-4. The Contractor **shall** provide the personnel, facilities, and hardware necessary to prepare and pack the instrument and its GSE for shipment and **shall** be responsible for the transportation and shipment of the material to the designated spacecraft facility, or other required location. The Contractor **shall** prepare, pack, and ship all instrument subsystems, systems, and the simulators between the places of manufacture, and integration and test, and storage and delivery in appropriate shipment containers. The Contractor **shall** prepare, pack, and ship all related calibration, mechanical, and electrical GSE required to support any instrument subsystems, systems, and simulators during each phase of test, integration, and launch preparation.

PHS&T Plans (included in IT-4) **shall** include long term (10 years) storage plans and necessary storage related testing, cleaning, and maintenance for the ETU, FM1, and additional FMs as required.

The Contractor **shall** design, develop, and deliver shipping and storage containers and any shipping environment-related monitoring equipment and contamination witness plates necessary for the ETU and additional FMs as needed based on the IMS and the Contractor's I&T Plan.

Instrument EGSE **shall** make accessible instrument-relevant test data, such as chamber telemetry, analog test point, accelerometer, and thermocouple data. The content of this data will be determined jointly by the instrument and spacecraft providers and defined in the applicable instrument ICD.

The electrical interface between the spacecraft bus and the instrument **shall** be checked out prior to instrument delivery to the spacecraft using an electrical interface simulator provided by the spacecraft contractor.

6.3 Pre-Ship Review

Prior to shipment, the Contractor **shall** conduct a PSR in accordance with CDRL RE-7 or delta review. All documentation required to manufacture, test, calibrate, and deliver the hardware as well as all test data of the specific hardware under review **shall** be available for the Government's review at the PSRs.

6.4 Delivery, Checkout, and Acceptance

The Contractor **shall** unpack the flight hardware and GSE upon its arrival at the spacecraft contractor's facility, perform a visual inspection to detect physical damage, set up a bench test environment, and perform a post-delivery functional check out of the instrument.

The spacecraft contractor will provide facilities, security, storage, standard cleaning supplies, and standard test equipment (multi-meter, probes, grounding, ionized air, etc.). The Contractor **shall** provide a bench check-out test report including comparison to the identical pre-ship test performed at the Contractor's facility. The Contractor **shall** compare the instrument post-shipment functional test results with the pre-shipment functional test results and provide comparative analyses and results in accordance with CDRL IT-2.

Government conditional acceptance will occur after acceptance testing confirms successful completion of mechanical and electrical integration of the instrument with the spacecraft. The Contractor **shall** be responsible for the instrument handling and operation from delivery until lifted "on-hook" by the spacecraft contractor for mechanical integration with the spacecraft. The spacecraft contractor will be responsible for maintaining and controlling a safe environment for all instrument hardware and personnel from the time of arrival at the spacecraft contractor's facility until delivered to the launch site, or ready to ship for return to the Instrument Contractor, as directed by the Government.

6.4.1 Hardware Delivery Documentation

6.4.1.1 Operations Support Manuals

The Contractor **shall** provide the following Operations Support Manuals to maintain and monitor the instrument while in storage, and to operate it during satellite integration and test and on-orbit:

- a) The Instrument On-Orbit Commissioning Plan, CDRL OO-1.
- b) The Instrument User's Manual, CDRL OO-2
- c) The Instrument Command and Telemetry Handbook and C&T Database, CDRL OO-4.
- d) Instrument On-Orbit Launch and On-Orbit Operations and Contingency Procedures, CDRL OO-5.

6.4.1.2 Technical Manuals

The Contractor **shall** provide the following documents with the auxiliary information needed for the operation of the instrument:

- a) The Instrument User's Manual, CDRL OO-2.

6.4.2 Documentation Delivered with Hardware

Handling instructions/procedures and accompanying documentation may be inside the outer shipment package but **shall** be outside the sealed inner package and be accessible upon opening of the outer box, without having to remove the inner box.

6.4.3 Photographic Documentation

The Contractor **shall** produce, archive, and make available to the Government upon request, closeout photographic documentation of all assemblies during the manufacturing process and of the final integrated configuration "as flown." (Include name, number, date, assembly/sub-assembly, etc.)

Following the MRR, the Contractor **shall** provide a small number of photos each month for public release.

6.4.4 Test Data Files

The Contractor **shall** provide interim data files of the test data taken during the instrument test sequence in the Contractor's format and with appropriate logs in a timely manner for real-time discussion of test results with the Government upon request. The Government will provide access to an approved electronic document repository (e.g., the external SharePoint where test data will be posted).

6.5 Sensor Database Support

The Contractor **shall** supply XML (TBR) formatted database inputs as required by the ground system.

6.5.1 Sensor Command and Telemetry Database Support

The Contractor **shall** deliver the Instrument Command and Telemetry Database (CTDB) (CDRL OO-4). The Contractor **shall** place the CTDB under formal CM. In addition, the Contractor **shall** prepare and deliver a CTDB Validation Plan (CDRL OO-7) that describes the process used to verify and validate:

- a. The instrument-level database (CDRL OO-4).
- b. The system-level database.

6.5.2 Instrument Mission Data Structure Definition Support

The Contractor **shall** deliver the Instrument Mission Data Definitions (CDRL OO-4) to be used within the Missions Operation Element to identify and process mission data produced by the sensor.

6.5.3 Instrument Memory Table Support

The Contractor **shall** deliver the Instrument Memory Table Structure Definitions and baseline content for the loadable and dumpable Instrument Memory Tables, to be used to populate the Ground Reference Image, in accordance with CDRL OO-4.

6.6 Instrument Post-Delivery Support

This section describes the Contractor's staffing and material responsibilities to support the mission system implementation, which includes satellite integration and test, training, mission readiness support, launch site support, and support through and beyond operations hand-over through the life of the Contract. The Contractor **shall** be responsible for the operations support and performance evaluation of each instrument through satellite encapsulation.

The post-delivery period (from delivery of the SMBA through on-orbit support) is divided into the following sub-periods for the purposes of explicitly defining Contractor support:

- PDSP = Activities prior to delivery.
- PDS = The period from delivery to Launch.
- Commissioning Support = The period from Launch to Launch + 90 days.
- Sustaining Engineering and Cal/Val Support.

- The Contractor **shall** support the Cal/Val Planning Process as an active member of the Cal/Val Planning Team.
- Post Launch Support = The period beyond Launch + 6 months, through the duration of the Contract. (Performed under task orders issued against CLIN 0003, defined in Contract clauses GSFC 52.211-91, SCOPE OF WORK (FEB 2016), and 1852.217-72, PHASED ACQUISITION USING PROGRESSIVE COMPETITION DOWN-SELECTION PROCEDURES.

The instrument Contractor **shall** provide satellite I&T, mission operations readiness support and training, launch site support, and support through operations hand-over in accordance with CDRL OO-5, Instrument Launch and On-Orbit Operations and Contingency Procedures.

The instrument Contractor **shall** provide the following inputs to the spacecraft contractor for Instrument GSE and Flight Hardware testing as necessary as part of risk reduction testing and/or as part of satellite acceptance testing:

- Applicable instrument procedures (CDRL IT-3) or inputs to spacecraft procedures that are necessary to run the GSE, ETU, or Flight Hardware.
 - The inputs to spacecraft procedures capturing all activities, equipment, personnel, and support required for the Contractor's instrument to be ready for flight after being delivered to the spacecraft contractor **shall** be documented through a PDS document in accordance with CDRL IT-4.
- Applicable instrument scripts/software or inputs to spacecraft scripts/software that are necessary to run the GSE, ETU, or Flight Hardware (CDRL IT-4).

The Government's intent is to do this in a collaborative environment between the instrument Contractor, the spacecraft contractor, and the Government.

6.6.1 General Roles and Responsibilities (PDSP and PDS Phases)

Prior to launch, the Contractor **shall** be the sole responsible party for mechanical instrument support, including, but not limited to, red and green tag removal and installation, cleaning, inspection, etc. The Contractor **shall** be responsible for the operation and physical handling of the instrument until the spacecraft contractor accepts it for mechanical integration with the spacecraft, and whenever the instrument is de-integrated from the spacecraft. The Contractor **shall** be responsible for operating and maintaining their GSE at the spacecraft contractor and launch site facilities. The Contractor **shall** be responsible for shipping the instrument GSE from the launch site payload processing facility in a timely manner after launch. The Contractor **shall** be responsible for monitoring and reporting the instrument health and safety at all times the instrument is powered. The Contractor **shall** have personnel on-site during powered operations.

6.6.2 System Integration and Test Support

6.6.2.1 Instrument Contractor Personnel and Facility Support

The Contractor **shall** provide personnel necessary to support integration and test activities at both the Contractor's facility and the spacecraft integration facility, including: performing instrument bench checkout; providing technical support during integration of the instrument to the spacecraft; performing all required repairs and re-tests; supporting limited and comprehensive performance testing throughout satellite qualification; performing thermal, mechanical, and electrical close-out; cleaning and monitoring contamination as needed; and supporting electrical

system checkout following integration. The Contractor **shall** provide the survival heaters that will be powered by the spacecraft.

To prepare for the activities listed above, the Contractor **shall** be required to support periodic virtual and in person face-to-face TIMs with NASA and the spacecraft contractor. These TIMs will be used to define satellite Limited Performance Tests, satellite CPTs, satellite environmental tests, and miscellaneous test details.

6.6.2.2 Instrument Performance Evaluation

The Contractor **shall** support instrument performance evaluation (e.g., data reduction and trending) during all phases of satellite integration and testing, including, test support, anomaly resolution support, and instrument data archival at the Contractor's facility, with electronic access provided to the Government, as required. Data resulting from these evaluations and tools utilized by the Contractor **shall** be made available upon request by the Government.

6.6.2.3 Instrument Data Products Support

The instrument Contractor **shall** provide the necessary instrument operating and test procedures (CDRL IT-3), parameters, screen displays, databases (CDRL OO-4), red and yellow limits definition (CDRLs OO-4), and instrument Constraints, Restrictions, and Warnings/Alerts (Instrument ICD CDRL SE-8). The instrument Contractor **shall** provide support for the conversion and validation of these data. This support **shall** include as a minimum a review of the converted documentation (page display printouts, procedure printouts) as well as validation in a satellite test environment. Additionally, the instrument Contractor **shall** provide instrument safety information and may be required to support periodic safety meetings. Instrument Data Sets for use in Ground Segment testing **shall** be provided in accordance with CDRL SW-5.

The Contractor **shall** prepare and accommodate remote network access from spacecraft contractor facility to the instrument vendor facility in order to transmit test data efficiently to the instrument vendor facility for off-line analysis.

6.6.2.4 Interface Testing

The Contractor **shall** support interface testing between their instrument (or a reasonable facsimile) and the spacecraft C&DH system (or a reasonable facsimile), including interfacing and testing the instrument (simulators, Engineering Development Unit, Flight Module, or other) and instrument GSE with the spacecraft data systems and C&DH simulators (including, but not limited to, the SIIS) to demonstrate proper operation. In support of these activities, the Contractor **shall** maintain, operate, and deliver instrument GSE hardware and software to the interface testing facilities.

As part of the above defined interface testing, the Contractor **shall** plan and implement support of spacecraft to payload interface electronics development risk mitigation testing. Technical support will be provided by the Contractor at the spacecraft contractor facility. This will support early risk reduction of instrument-spacecraft interfaces.

As part of the above defined interface testing, the Contractor (in conjunction with the spacecraft contractor) **shall** plan and implement the testing of instrument-spacecraft interfaces with the SIIS provided by the spacecraft contractor. Technical support will be provided by the spacecraft

contractor at the instrument simulator provider facility and any training necessary to facilitate such support by the spacecraft contractor will be provided by the instrument simulator provider. This will support early risk reduction of instrument-spacecraft interfaces.

6.6.2.5 Engineering Test Unit Support

The goal of the ETU phase of satellite I&T is to perform instrument mechanical and electrical integration; instrument power, command, telemetry, and data interface verification; and procedure and database validation. This may include thermal testing at ambient pressure and/or vacuum and EMI testing. The Contractor **shall** support ETU efforts related to risk reduction for flight unit integration and test.

Additional risk reduction activities utilizing the SMBA ETU at the spacecraft contractor facility include, but are not limited to:

- SIIS testing with SMBA ETU.
- Inter-Operability Testing with a representative SMBA unit.
- EMI risk reduction testing with SMBA ETU.

6.6.2.6 Bench Checkout

The Contractor **shall** provide technical support for the instrument at the spacecraft facility. The goal of the bench checkout at the spacecraft facility is to certify that the instrument has survived shipment to the spacecraft facility by replicating the pre-ship limited functional test. The Contractor **shall** define a limited functional test for the instrument, using the instrument GSE, for use after delivery to the spacecraft contractor's facility. The Contractor **shall** be responsible for performing this test prior to integrating the instrument to spacecraft. The instrument EGSE **shall** be used to perform the stand-alone Bench Acceptance Test when the instrument arrives at the spacecraft contractor.

6.6.2.7 Flight Model Support

The Contractor **shall** provide a FM instrument, with necessary GSE and personnel, to support integration and testing with the spacecraft. Satellite integration and test will encompass mechanical integration (including alignment), electrical integration, limited and comprehensive performance testing at ambient, satellite environmental qualification, and post-environmental functional and performance testing. The Contractor **shall** support the satellite Pre-Environmental Review held prior to satellite environmental testing, and the satellite Pre-Ship Review held prior to shipment to the launch site.

6.6.3 SMBA OpSim Support

The Contractor **shall** provide PDS for the delivered simulators.

6.6.4 Launch Site Support

The Contractor **shall** provide launch site support at the payload processing facility as required including the following:

- Limited satellite functional testing prior to integration to the launch vehicle.
- Final thermal, electrical and mechanical closeout of the instrument.
- Final red tag/green tag removal and installation.
- Final instrument inspection and cleaning.
- Anomaly support.

The Contractor **shall** support Flight Readiness Review, Launch Readiness Review, and other reviews as required.

6.6.5 Mission System Integration Support

The Contractor **shall** support planning and execution of mission systems integration activities through satellite operational hand-over, including training and mission readiness support as described below.

6.6.5.1 Integrated Mission Timeline

The Contractor **shall** support development of Integrated Mission Timeline (IMT), including:

- a) Support for regular IMT Working Group.
- b) Review IMT prior to major stakeholder reviews and attend reviews.

6.6.5.2 Operations Reviews

The Contractor **shall** support operations reviews including but not limited to MOR, ORR, Mission Readiness Review, and Operations Hand-over Review, specifically:

- a) Review instrument-specific information in review materials for accuracy and completeness.
- b) Provide Q&A support during instrument-specific sections of review presentations.

6.6.5.3 Rehearsal Anomaly Definition

The Contractor **shall** support Rehearsal Anomaly Definition, including:

- a) Defining anomaly scenarios to inject into rehearsals.
- b) Assist in scripting anomalies for the simulator.

6.6.5.4 Satellite Operations Exercises

The Contractor **shall** participate in Satellite Operations Exercises (SOE), including:

- a) Participate in planning of instrument-focused SOEs.
- b) Review and Approve Test Plans, Procedures, and Configurations.
- c) Support Test Readiness Review.
- d) Monitor execution of instrument-focused SOEs at Spacecraft Facility and the NOAA Satellite Operations Facility.

6.6.5.5 Compatibility Tests

The Contractor **shall** participate in satellite-level compatibility tests per instrument, including:

- a) Participate in planning of instrument-focused compatibility.
- b) Review and Approve Test Plans, Procedures, and Configurations.
- c) Support Readiness Review.
- d) Participate in Early Engineering Opportunity, dry-run, and run-for-record execution of tests.

6.6.5.6 Mission Rehearsals

The Contractor **shall** participate in Mission Rehearsals, including:

- a) Three (3) (TBR) instrument-focused rehearsals and dress rehearsals.
- b) Three (3) (TBR) Support Operations Readiness Exercises as needed to prepare for and/or supplement Mission Rehearsals.

6.6.5.7 Operations Product Validation

The Contractor **shall** support Operations Product Validation, including:

- a) Review PROCs for consistency with ROPs.
- b) Review other ops products such as pages and tasks for accuracy and contingency procedures.
- c) Witness execution of final PROCs to assess readiness for flight.

6.6.5.8 Recommended Operating Procedures

The Contractor **shall** develop Instrument ROPs (OO-5) and support:

- a) Recommend ROP changes as needed based on lessons learned.
- b) Review any Mission Operations Team ROP changes based on operational lessons learned.
- c) Update PROCs as needed.

6.6.5.9 Operational Readiness Test

The Contractor **shall** support Operational Readiness Test (ORT) that includes the review of instrument-specific information in ORT Plan for accuracy and completeness.

6.6.6 Commissioning Support

"The Contractor **shall** perform all of Section 6.6.6 as core requirements with the exception of the task order requirements in CLIN 0003, defined in contract clauses GSFC 52.211-91, SCOPE OF WORK (FEB 2016), and 1852.217-72, PHASED ACQUISITION USING PROGRESSIVE COMPETITION DOWN-SELECTION PROCEDURES."

The Contractor **shall** support Satellite Launch, Early Orbit, and Anomaly checkout, including providing necessary engineering support at the operations facility for on-orbit activation and checkout of the instrument. The Contractor **shall** provide at a minimum:

- Instrument support during bus commissioning at the operations facility (monitoring instrument survival telemetry, spacecraft out-gassing, and other instrument concerns as needed). Commissioning reports **shall** be provided in accordance with CDRL OO-3.
- Instrument participation in real-time go/no-go decision-making during instrument activation at the operations facility.
- Support during buy-off.
- Anomaly resolution support.
- Support during instrument calibration and validation.
- Verification of SMBA health, functionality, and performance. Evaluations to include trending and comparisons to requirements and expectations established during ground test and characterizations. The results of all evaluations **shall** be documented in Commissioning Report (CDRL OO-3).

6.6.7 Post-Launch Sustaining Engineering

"The Contractor **shall** perform all of Section 6.6.7 as core requirements with the exception of the task order requirements in CLIN 0003, defined in contract clauses GSFC 52.211-91, SCOPE OF WORK (FEB 2016), and 1852.217-72, PHASED ACQUISITION USING PROGRESSIVE COMPETITION DOWN-SELECTION PROCEDURES."

The Contractor **shall** support the mission by providing:

- Anomaly resolution support through the period of performance of the Contract.
- Software maintenance through the period of performance of the Contract (includes Flight, Simulator, and GSE software).

Appendix A ABBREVIATIONS AND ACRONYMS

Abbreviation/Acronym	Definition
ARB	Anomaly Review Board
ANSI	American National Standards Institute
ASNT	American Society for Nondestructive Testing
ATMS	Advanced Technology Microwave Sounder
BCR	Baseline Change Request
BOL	Beginning of Life
C&DH	Command and Data Handling
Cal/Val	Calibration/Validation
CCB	Configuration Control Board
CCP	Contamination Control Plan
CCR	Configuration Change Request
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CIL	Critical Items List
CLIN	Contract Line Item Number
CM	Configuration Management
COR	Contracting Officer's Representative
COTS	Commercial-off-the-Shelf
CPT	Comprehensive Performance Test
CRAE	Commodity Risk Assessment Engineer
CSO	Chief Safety and Mission Assurance Officer
CTB	Consent to Break
CTDB	Command and Telemetry Database
CTE	Calibration Test Equipment
CTP	Consent to Proceed
CTI	Consent to Integrate
DID	Data Item Description
DMR	Design Modification Review
EEE	Electrical, Electronic, and Electromechanical
EGSE	Electrical Ground Support Equipment
EIADP	End Item Acceptance Data Package
EIDP	End Item Data Package
EMC	Electromagnetic Capability
EMI	Electromagnetic Interference
EOL	End of Life
EPR	Engineering Peer Review
ETU	Engineering Test Unit

EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EVM(S)	Earned Value Management (System)
FAR	Federal Acquisition Requirements
FM	Flight Model
FOD	Foreign Object Debris
FPGA	Field Programmable Gate Array
FRB	Failure Review Board
FSW	Flight Software
FSWDE	Flight Software Development Environment
FVTS	Flight Vehicle Test Suite
GCT	Ground Compatibility Test
GEVS	GSFC Environmental Verification Specifications
GIDEP	Government-Industry Data Exchange Program
GMIP	Government Mandatory Inspection Point
GOTS	Government off-the-shelf software
GOLD	GSFC Open Learning Design
GOTS	Government-off-the-Shelf
GPR	Goddard Procedural Requirement
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
G-R	Giver/Receiver
Hz	Hertz
I&T	Integration and Test
IASI	Infrared Atmospheric Sounding Interferometer
IAW	In Accordance With
IBR	Integrated Baseline Review
ICD	Interface Control Document
IIRA	Inherited Item Risk Assessment
IM	Instrument Manager
IMAR	Instrument Mission Assurance Requirements
IMS	Integrated Master Schedule
IMT	Integrated Mission Timeline
IPMDAR	Integrated Program Management Data Analysis Report
IRCD	Interface Requirements Control Document
IRD	Interface Requirement Document
ISAR	Instrument Safety Assessment Report
IV&V	Independent Verification and Validation
JPSS	Joint Polar Satellite System
K	Kelvin

LEO	Low Earth Orbit
LPD	Low Earth Orbit Programs Division
LPT	Limited Performance Testing
LTAN	Local Time of Ascending Node
LTS	Lower-Tier Supplier
M&P	Materials and Processes
MA	Mission Assurance
MAR	Mission Assurance Requirements
MGSE	Mechanical Ground Support Equipment
MICD	Mechanical Interface Control Document
MOR	Mission Operations Review
MOTS	Modified-off-the-Shelf
MRB	Material Review Board
MRR	Manufacturing Readiness Review
MWS	Microwave Sounder
NASA	National Aeronautics and Space Administration
NEdT	Noise Equivalent Differential Temperature
NEON	Near Earth Orbit Network
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NPSD	Noise Power Spectral Density
NS1	NEON Series-1
NSOSA	NOAA Satellite Observation System Architecture
OBS	Organization Breakdown Structure
OpSim	Operations Simulator
ORR	Operational Readiness Review
ORT	Operational Readiness Test
PCB	Parts Control Board
PCB	Printed Circuit Board
PDR	Preliminary Design Review
PDS	Post Delivery Support
PDSP	Post Delivery Support Planning
PER	Pre-Environmental Review
PHS&T	Packaging, Handling, Storage, and Transportation
PM	Program Manager
PMB	Performance Measurement Baseline
PMP	Project Management Plan
PMR	Project Management Review
PRD	Performance Requirements Document

PROC	Procedure
PSR	Pre-Ship Review
q	Moisture
RDR	Raw Data Record
RFI	Radio Frequency Interference
RMP	Risk Management Plan
ROP	Recommended Operating Procedure
RTM	Reduced Thermal Model
SAE	Society of Automotive Engineers
SDF	Software Development Folder
SDP	Software Development Plan
SDR	System Definition Review
SEN	Software Engineering Notebook
SIIS	Spacecraft-Instrument Interface Simulator
SMA	Safety and Mission Assurance
SMBA	Sounder for Microwave-Based Applications
SOE	Satellite Operations Exercise
SOW	Statement of Work
SPF	Single Point Failure
SPMM	Sensor Performance Math Model
SRF	Spectral Response Function
SRR	Systems Requirements Review
SRS	Software Requirements Specification
SSO	Sun-Synchronous Orbit
STE	System Test Equipment
SW	Software
SWAR	Software Acceptance Review
T	Temperature
TBD	To Be Determined
TBP	To Be Proposed
TBR	To Be Reviewed
TBS	To Be Supplied
TDR	Test Data Review
TIM	Technical Interchange Meeting
TPM	Technical Performance Metrics
TRR	Test Readiness Review
TVAC	Thermal Vacuum
V&V	Verification and Validation